

12.1.1.1 Prerequisites for use

A valid Software Maintenance Contract must exist in order to use the EPLAN Data Portal. The EPLAN Data Portal can then be used as desired by providing data from the contract. EPLAN distinguishes specific user groups on the basis of their Software Maintenance Contract levels, but this does not play a major role in the use of the EPLAN Data Portal. As always, the more comprehensive and more precise the Software Maintenance Contract, the greater the number of options are for using the EPLAN Data Portal. The following sections will be limited to standard use.



NOTE: The EPLAN Data Portal is automatically installed when, for example, EPLAN Electric P8 is installed. Separate installation is not necessary.

12.1.2 Before the first start

An account must be created in order to use the EPLAN Data Portal. To do this, you start the **OPTIONS/SETTINGS/USER/MANAGEMENT/DATA PORTAL** menu item.

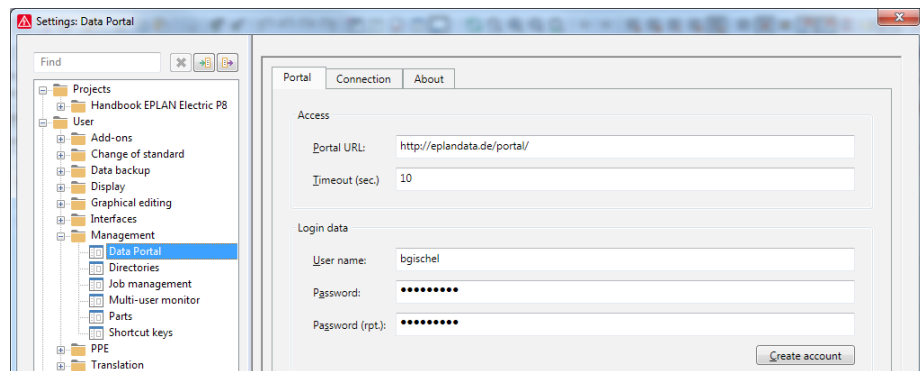


Fig. 12.1 EPLAN Data Portal settings

The settings for the EPLAN Data Portal are subdivided into several tabs. **Access** and **login data** are defined in the *Portal* tab (default setting already entered by EPLAN — does not need to be changed). This creates an account, and the user can use the Data Portal after completing the prompts.

If you do not have your own account yet, you must create it once. Define a user name and password, then re-enter the password a second time to confirm. Then, click the *Create account* button, and EPLAN will create the account.

The settings on the *Connection* tab normally do not need to be changed.

12.1.2.1 Data Portal navigator

The **Data Portal navigator** is the main focus of the EPLAN Data Portal. It is started via the UTILITIES/DATA PORTAL menu.

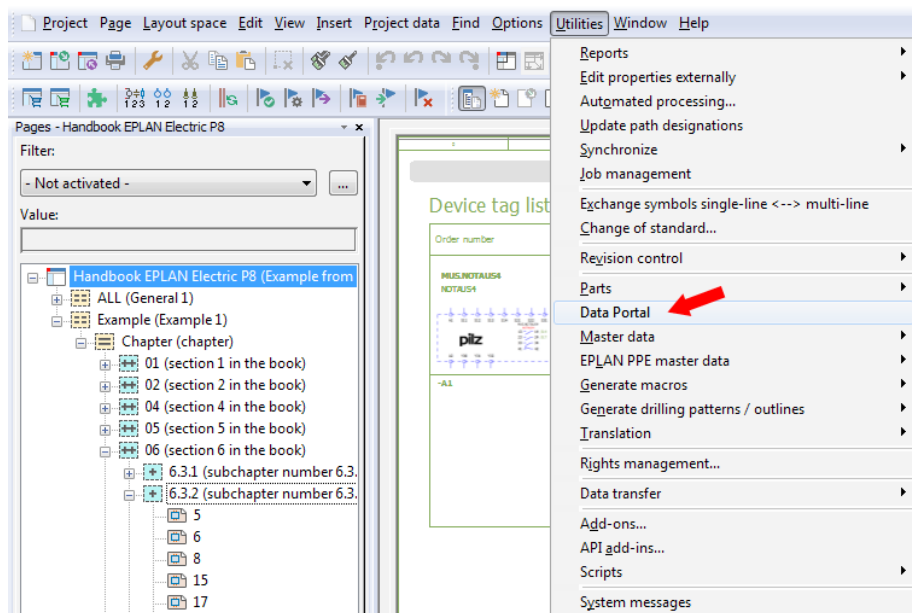


Fig. 12.2
Data Portal navigator
menu item

Like elsewhere in EPLAN, the navigator can be docked and undocked anywhere. This means that it can be freely placed on the desktop or docked to any desired position.

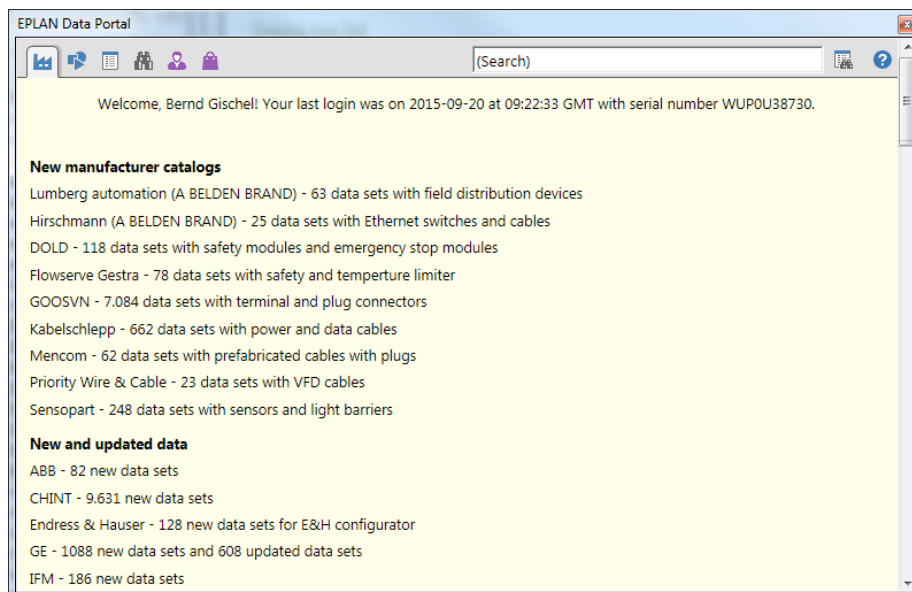


Fig. 12.3
Possible Data Portal
view with optional
information

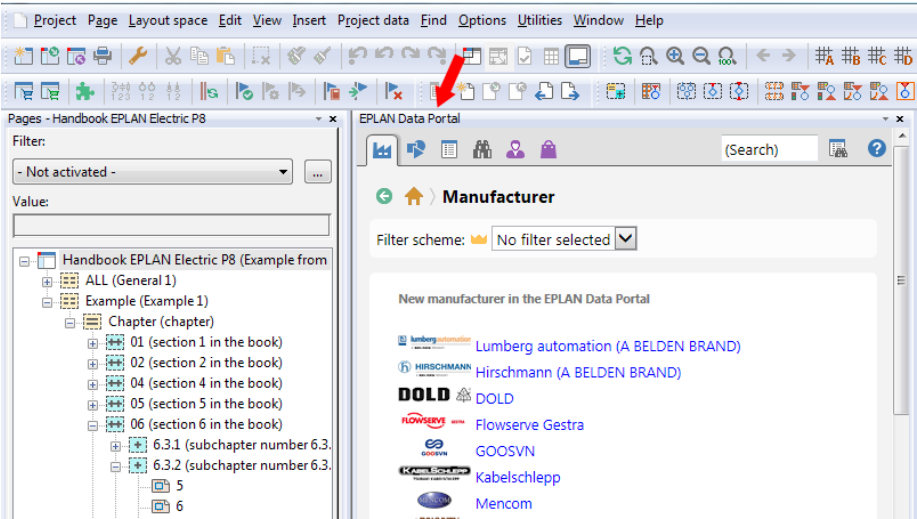


Fig. 12.4 The Data Portal navigator when docked

The navigation/toolbar, with buttons for functions such as **DISPLAY MANUFACTURERS**, **SEARCH**, **SETTINGS** and **ONLINE HELP**, is located under the navigator dialog logo.

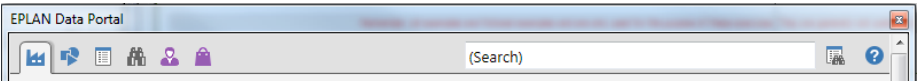


Fig. 12.5 The data portal navigator’s navigation/toolbar

The navigation bar buttons have the following meanings:

	Open start page		Display manufacturer
	Display catalog		Display list
	Display search with search criterion		Display user data
	Display shopping cart		Display help

Located a little bit “apart” from the other buttons is the  button that takes you back to the previously displayed page in the navigator.

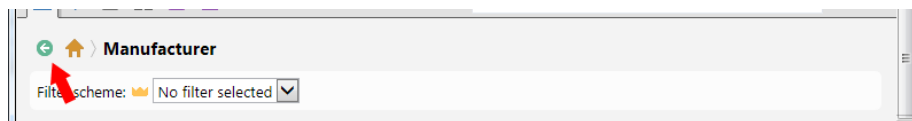


Fig. 12.6 Back button

Below the navigation/toolbar is the navigator's actual "data area". The first time the navigator is started, a list of the manufacturers in the EPLAN Data Portal is displayed here. You can click each manufacturer. The Portal opens the corresponding pages with the current manufacturer data. You can continue to navigate through this data.

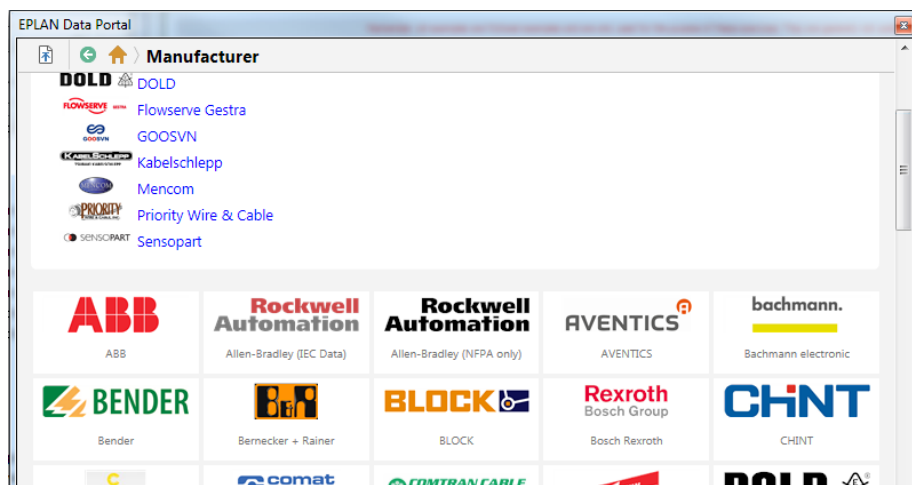


Fig. 12.7 Manufacturers in the Data Portal

12.1.3 How the EPLAN Data Portal works

How does the EPLAN Data Portal work? It is very simple. While you are designing, you can use the Portal to import or insert missing parts data, macros or a device list of the parts contained in the shopping cart onto schematic pages and/or parts management. This, of course, assumes that the missing data is actually available in the Data Portal.

12.1.3.1 Insert parts, macros and data from the Portal

A brief example will show how you can find a PNOZ device in the Portal and then insert it onto an open schematic page.

Requirement: The Data Portal navigator is started. Then, you start the search.

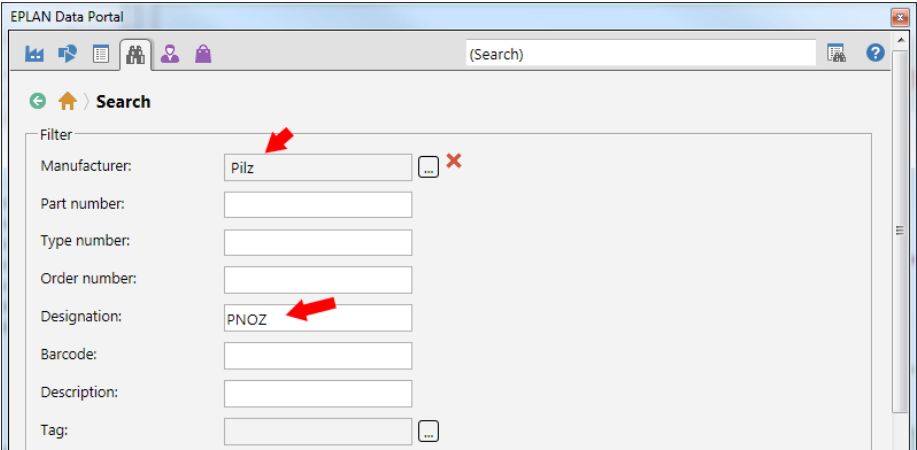


Fig. 12.8 Search function with criteria selection

EPLAN opens the **Search** dialog. Here you can enter in the **Designation** field the *PNOZ* value and, in the **Manufacturer** field, select the *Pilz* entry (via the *More* button) from the list of manufacturers. All other fields are not used and/or remain empty. Click the **FIND** button at the very bottom of the **Search** dialog to start the search.

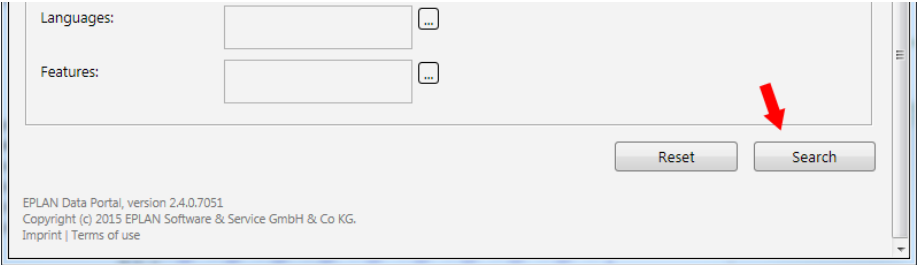


Fig. 12.9 Starting a search

If EPLAN finds parts matching the search term, the Data Portal lists these in an overview list. The list contains the parts found by the search as well as header data such as manufacturer, subgroup, languages and characteristics.

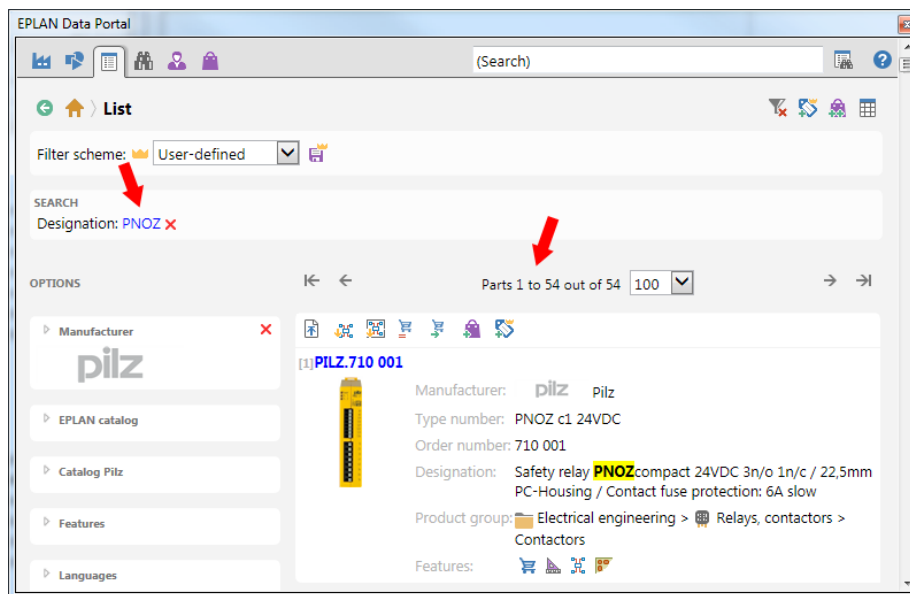


Fig. 12.10
Parts found

This list includes the devices, their part numbers, and other information such as type and order number, designation, the product group to which the device belongs, and what other characteristics the device possesses. Characteristics are additional information.

Simply stated, the more superior a part in the Portal is, the more characteristics it has in the Portal. Characteristics can be: The device has device data (function definitions/templates) or, for example, macros are available for the device.

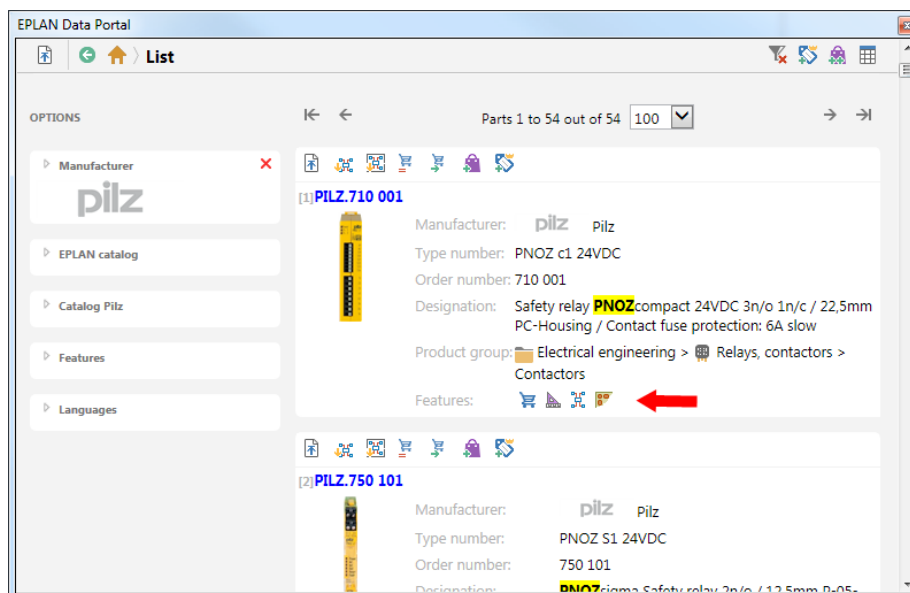


Fig. 12.11
Characteristics
of a part

 **NOTE:** The search results always depend on how the search and the various search criteria were set and also what data is actually available in the EPLAN Data Portal.

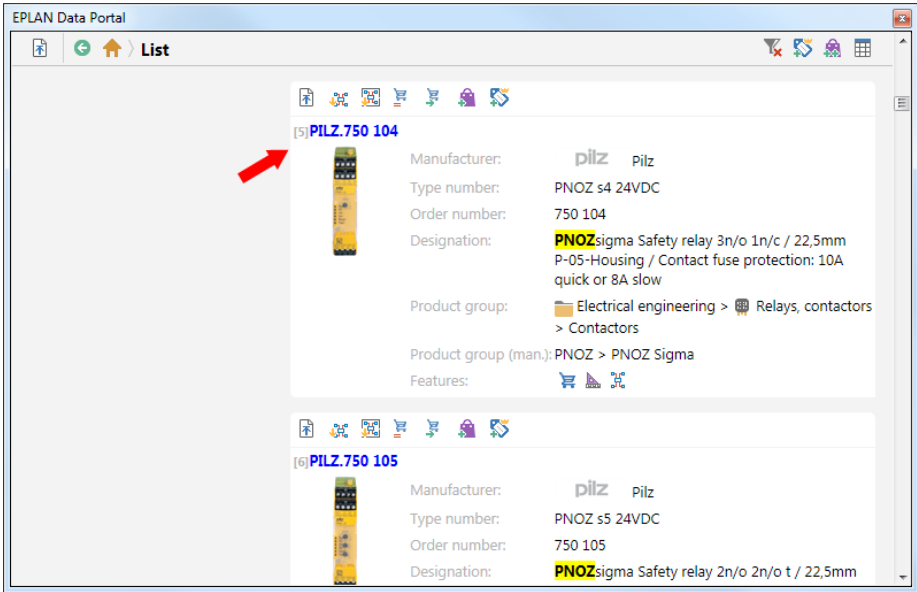


Fig. 12.12 Selected PNOZ device

In this example, we need part PILZ.750 104. This found device can now be directly used in the design.

The Portal list view has several buttons above the part that have the following meanings.

 Back to beginning of page

This button is self-explanatory. It takes you back to the beginning of the page of a filtered parts list.

 Insert macro <xyz> in graphical editor (on the schematic page)

If you click this button, EPLAN “hangs” the macro on the cursor, allowing it to be placed on the schematic page in the graphical editor.

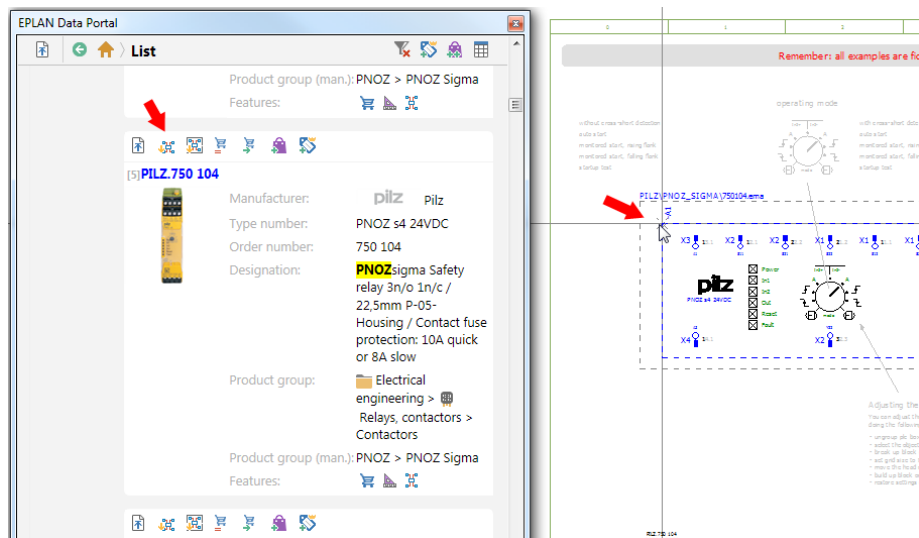
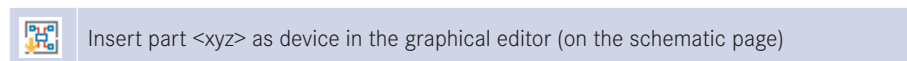


Fig. 12.13
Directly placing the
selected macro in
the graphical editor



If you activate this button, first the **IMPORT OF PARTS** prompt appears.

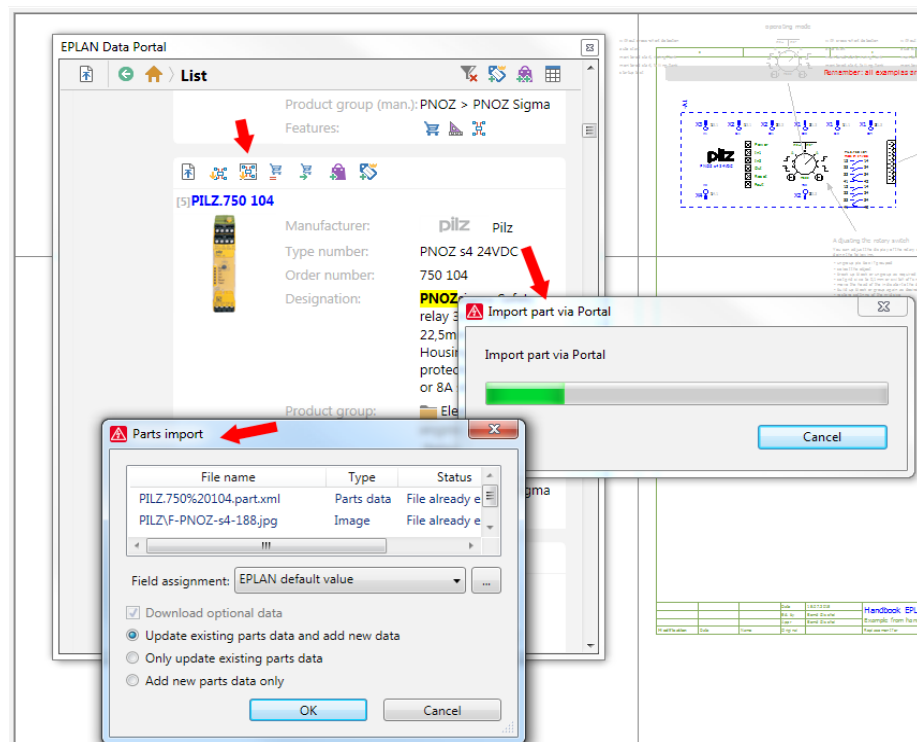


Fig. 12.14
Inserting the device in
the graphical editor

You need to decide how the data is to be stored in the existing **parts management** and whether, for example, the *optional data* should also be imported.

When you click OK in the **IMPORT OF PARTS** dialog, EPLAN imports the data into the **parts management** and then “hangs” the macro on the cursor. Now you can place the macro on the schematic page.



When this button is clicked, only the selected parts data is imported into the **parts management**.

EPLAN restarts the **IMPORT OF PARTS** dialog automatically and, depending on the settings, imports the corresponding parts data into parts management. The rest of the data is saved in the directory structure.

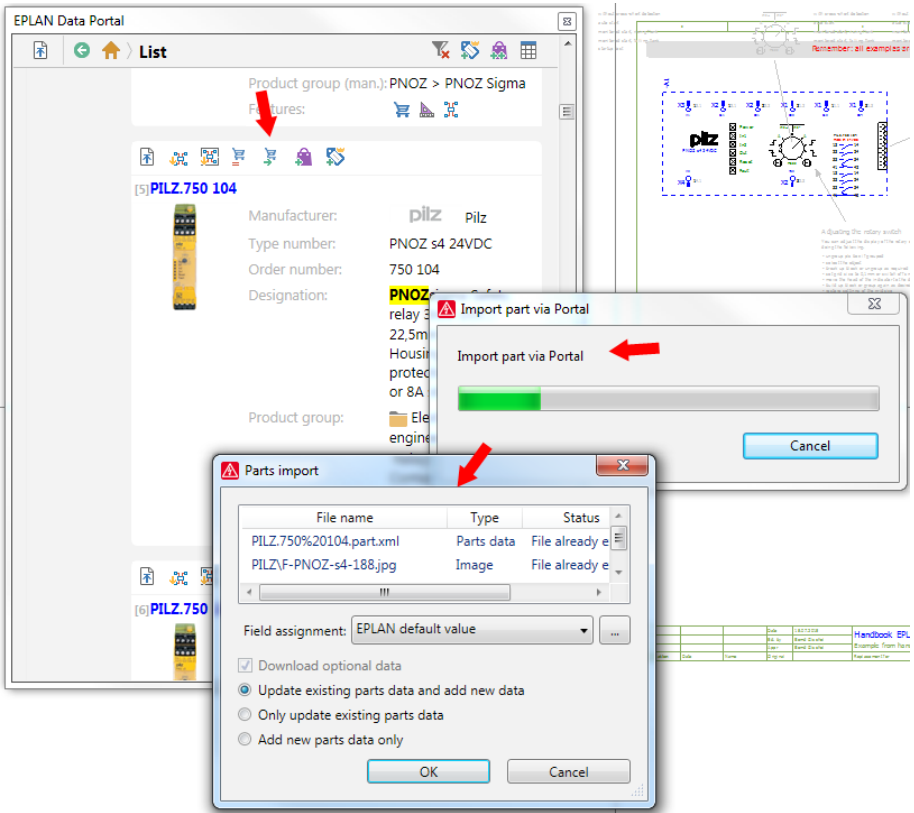


Fig. 12.15
Pure import of parts

After the import, EPLAN closes these dialogs and the graphical editor is active again.

 Put part <xyz> in the shopping cart (add)

Clicking this button adds the selected device to the shopping cart. The shopping cart is called up via the  button in the toolbar.

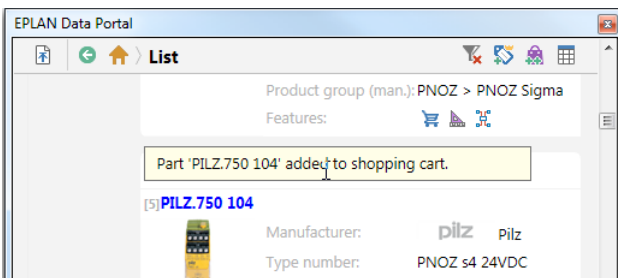


Fig. 12.16

Part added to shopping cart message

All devices in the shopping cart can be imported completely (collected).

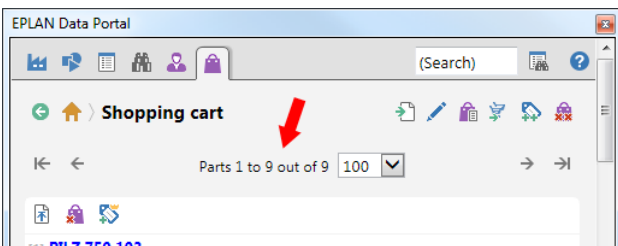


Fig. 12.17

Import the entire shopping cart

All of the parts can also be removed from the shopping cart.

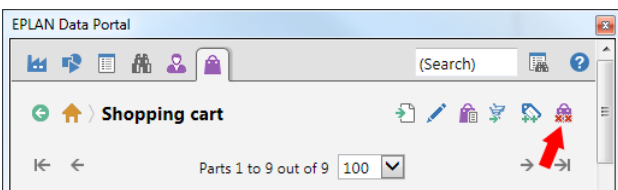


Fig. 12.18

Empty the entire shopping cart

 **NOTE:** Apart from inserting parts, you can also assign to the selected part parts data of already placed devices in the Data Portal.

 Assign parts data

12.1.3.2 Generate device lists

Apart from the functions described, such as the import of parts data and inserting of parts as macros in the graphical editor, EPLAN Data Portal offers the function that allows you to import as device lists into the current project parts currently in the shopping cart.

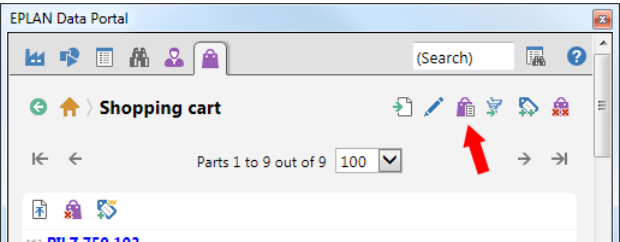


Fig. 12.19 Import shopping cart as device list button

Click the  button for this purpose. EPLAN now starts the import automatically.

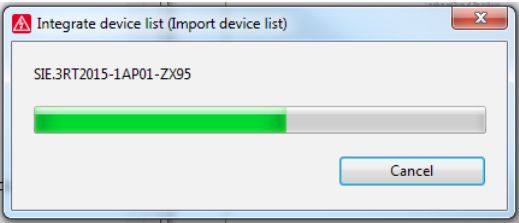


Fig. 12.20 Import device list dialog

When the import is complete, the device list will be available in the current project. The device list can be accessed via PROJECT DATA/DEVICES/PARTS/DEVICE LIST.

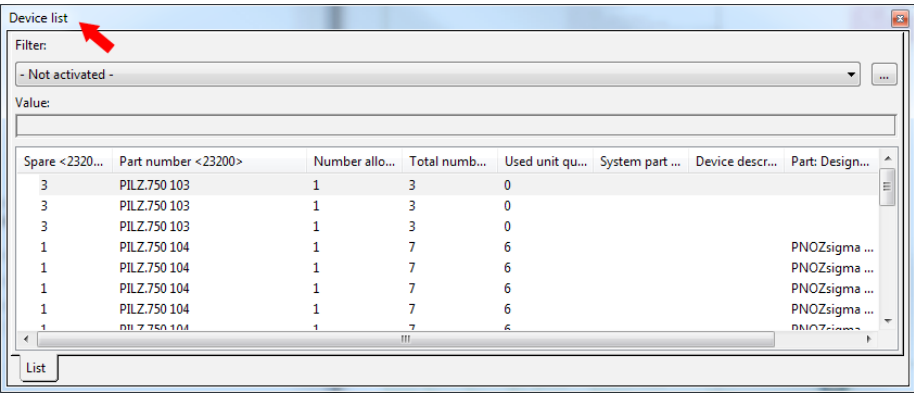


Fig. 12.21 Imported device list

■ 12.2 Project options

In EPLAN Electric P8, macros with value sets are very flexible, but they are limited in one aspect. Macros with value sets can only be used on the same page in EPLAN. Cross-page macros with value sets have not been available up to now. EPLAN closes this gap with the **Project options** module.

This section can only briefly touch on how to use project options, and only provides a short introduction to the procedure for creating project options. The large number of functions and the wide range of possibilities offered by the **Project options** module make it impossible to describe all the details of **project options** in the limited space available in this book. This section is intended therefore as food for thought, and to illustrate what is possible when using **project options**. The possible uses are (practically) limitless.

12.2.1 What are project options?

As the name indicates, project options are various options for (different design approaches) of a project. This can mean that, for example, the PLC in a project is implemented with a Siemens PLC for one customer and with a Schneider PLC for a different customer. This cannot be implemented easily using **VALUE SET MACROS**, but it is possible using the **Project options** module. You simply display or hide the desired options.

Project options cannot be created across projects. They are only available for the project for which they were created. It is possible to create project option templates that can be used in other projects.

12.2.2 Terminology in the Project options module

Project options are a type of “generic term”. To help to understand project options, the following section contains a brief explanation of the terms used.

- **Project options group:** This function allows grouping of project options. A project option group can contain several projects, but only one project option at a time can be switched on. All other project options in the project options group are then switched off. Project options groups can only be created in the **project options navigator**.
- **Project options:** These are partial areas of a project that can be switched on and off as desired. They can consist of one or more extracts, pages, page areas or unplaced objects. Project options also can only be created in the **project options navigator**.
- **Sections:** A section is always assigned to a project option. A section can consist of a partial circuit, a complete page or several pages. Unplaced objects are also possible in sections. In addition to using the **project option navigator**, sections can also be created from the **PROJECT DATA/PROJECT OPTIONS/CREATE SECTION** menu.

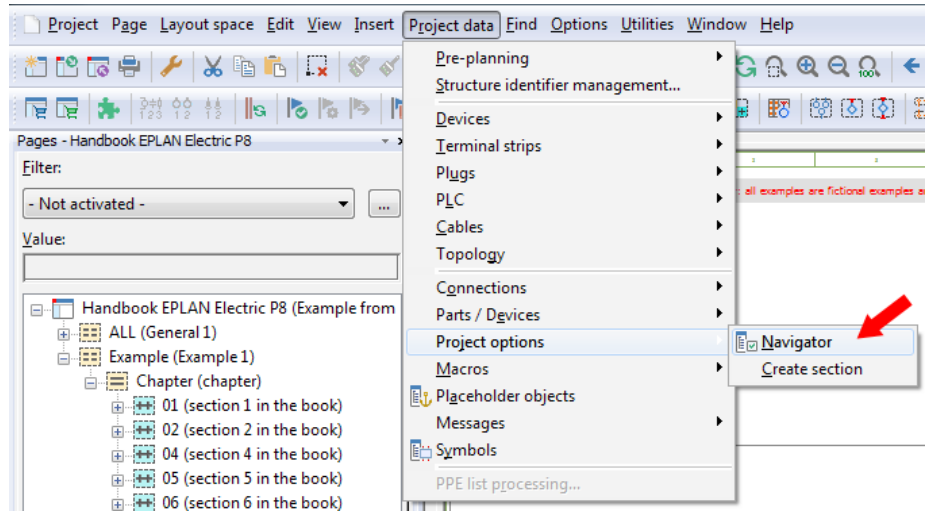


Fig. 12.22 Calling the project options navigator

- **Project options navigator:** The project options navigator is used for managing the project options groups, for creating and managing the project options, and switching the options on and off. It is called up via the **PROJECT DATA/PROJECT OPTIONS/NAVIGATOR** menu. The typical, familiar tree and list views are available within this navigator. Custom filters can also be set for the view in the navigator.

12.2.3 Creating options and sections

A bit of preparation is required for you to be able to easily “switch” between different options in a project. **Project options groups** and their affiliated **project options** must be created, and finally there is the most important point: creating the **sections**.

Section 12.2.3.1 explains in just a few steps the most important points for creating these options and their associated parts.

12.2.3.1 Project options group:

A **project options group** is created in the **project options navigator**. You start the navigator via the **PROJECT DATA/PROJECT OPTIONS/NAVIGATOR** menu.

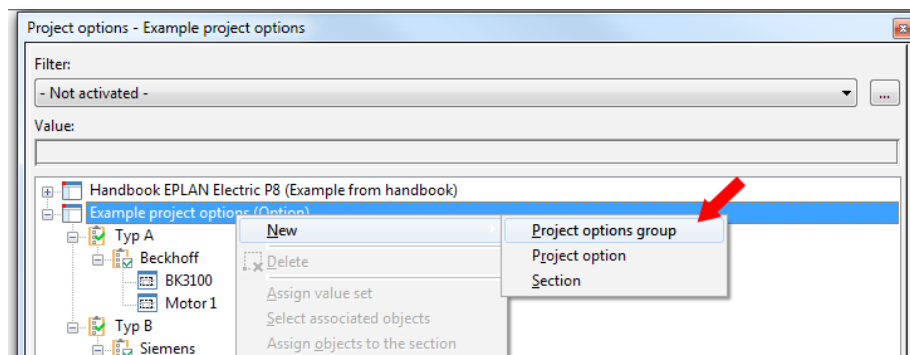


Fig. 12.23 Generating a project options group

Then you call up the **project options navigator** in the popup menu and select **NEW/PROJECT OPTIONS GROUP**. EPLAN opens the **PROJECT OPTIONS GROUP** dialog.

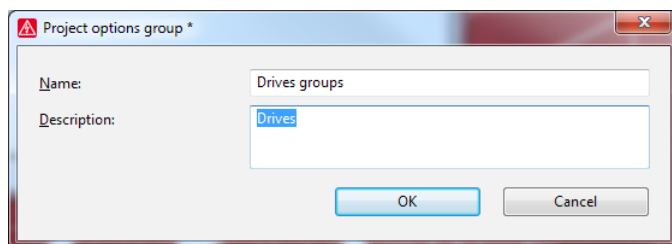


Fig. 12.24 Project options group dialog

You have to enter a **name** and **description**. When you click OK, the group is saved and displayed in the navigator.

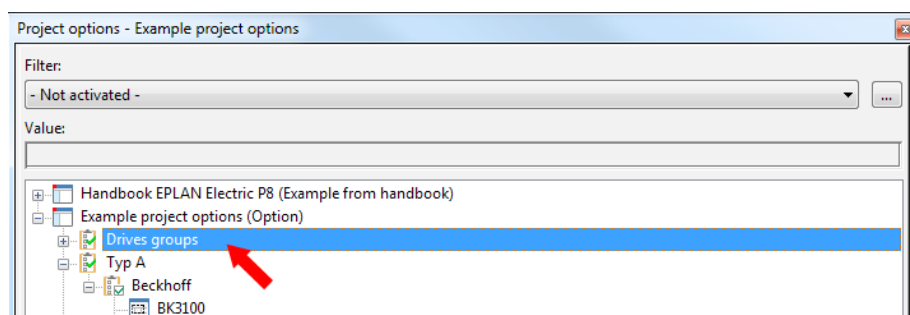


Fig. 12.25 View in the navigator

Examples of project options groups could be: *power supply*, *PLC*, *automatic operation*, *manual operation*, etc. The actual project options for power supply or PLC can be created beneath these project options groups.

12.2.3.2 Project options

Project options can also only be created in the **project options navigator**. Project options always belong to a project options group. This means that one of the project options has to be selected, or that a project options group must be selected when creating a project option. To do this, you open up the **project option navigator** and start the popup menu command **NEW/PROJECT OPTION**.

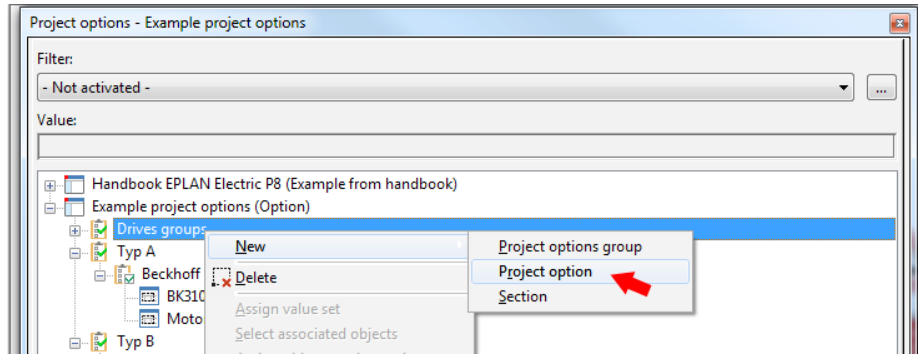


Fig. 12.26 Creating a new project option

EPLAN opens the **PROJECT OPTIONS GROUP** dialog. The project options group can be changed here, if desired. You have to assign the project option a **name** and can also enter a **description**.

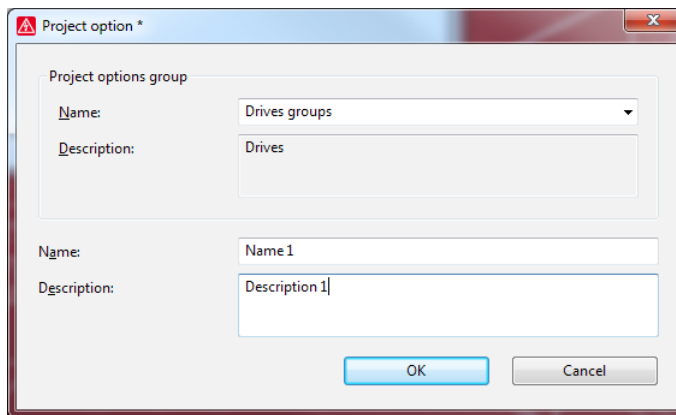


Fig. 12.27 Creating a new project option

When you click **OK**, the project option is saved and sorted under the selected project options group in the **project option navigator**.

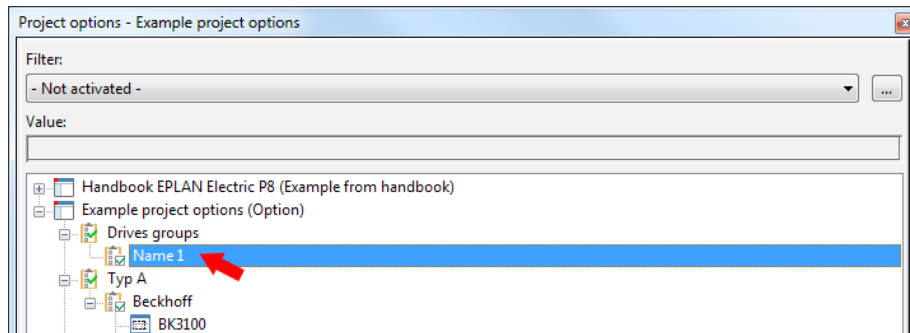


Fig. 12.28 Newly created project option

Several project options can be assigned to a project options group. However, only one project option at a time can be active, i.e. switched on.

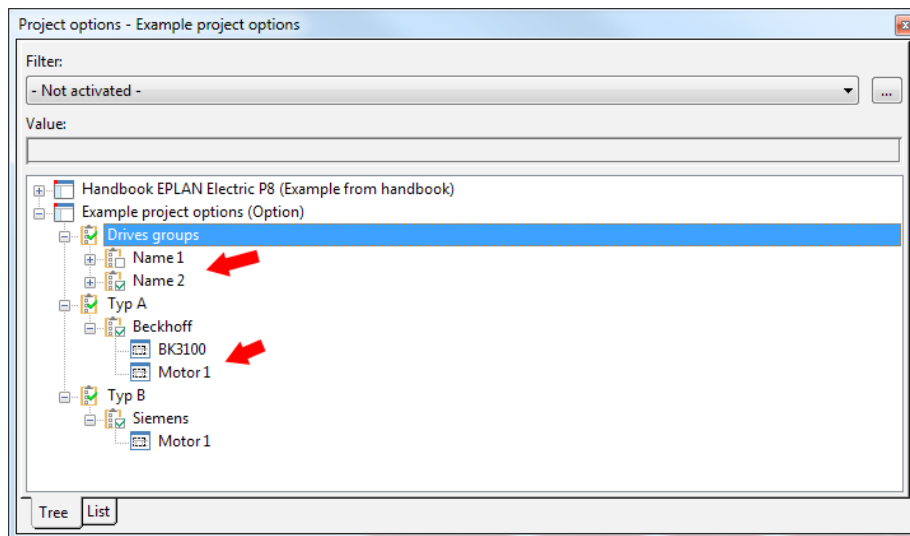


Fig. 12.29 Several project options

The  icon in front of the project option name indicates whether a project option is switched on. If it is switched on, you will see a small green check mark.

12.2.3.3 Sections

The actual **sections** can now be assigned within project options. As already described, sections can be partial sections, one or more pages, or unplaced functions.

To generate a section, in the **project options navigator**, you need to select the corresponding **project options group** and one of the **project options** below this group. Then you call up the popup menu and select **NEW/CREATE SECTION** to create the section.

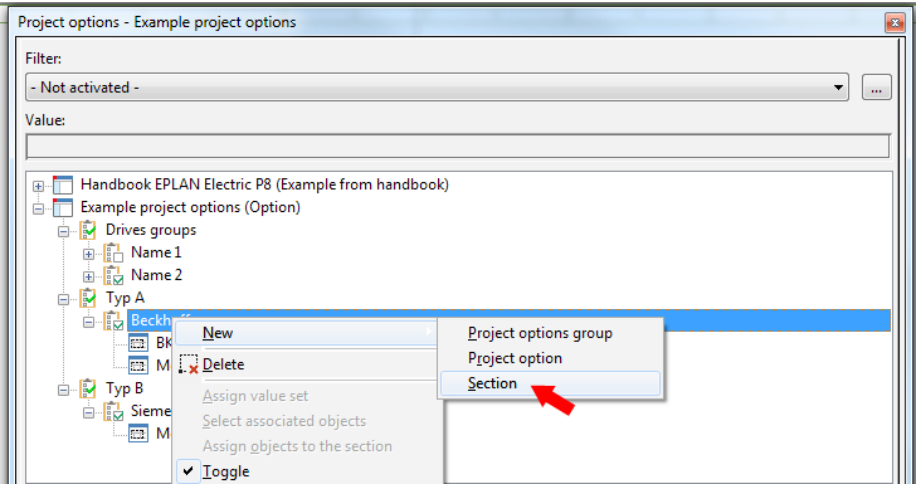


Fig. 12.30 Generating a new section

EPLAN starts the function. The objects belonging to this section must now be defined. This is done using windows. When you have finished defining the objects, you have to enter a **name** and **description** for the section in the subsequent **SECTION** dialog.

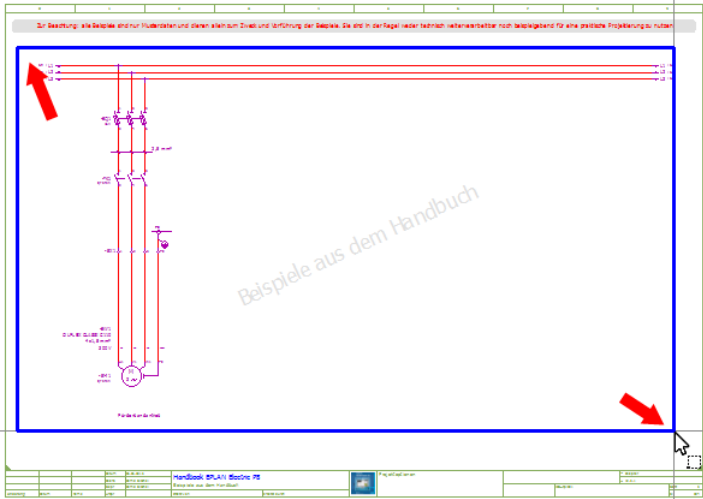


Fig. 12.31 Defining a section via a window

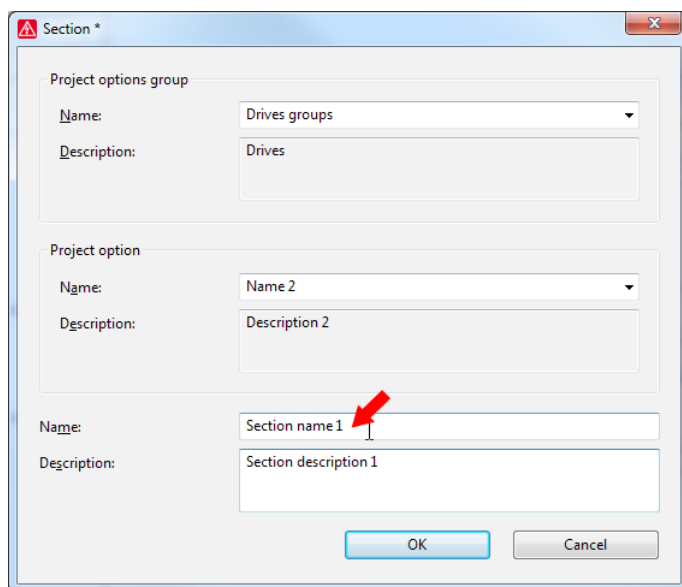


Fig. 12.32 Designating a new section

In the **SECTION** dialog, you once more have the opportunity to assign the section to a different project option or project options group. After you click the **OK** button, the section is saved and sorted under the associated **project options group** and **project option** in the **project options navigator**.

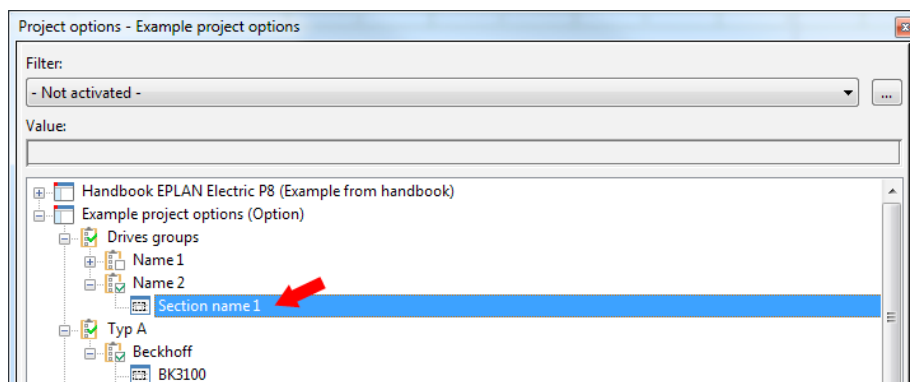


Fig. 12.33 Section in the navigator

Fig. 12.34 Editing new circuit parts for a new section

EPLAN visually switches the previous section to transparent. This is indicated by the yellow check mark  in front of the section name. The “new” section can now be created. You place and edit the desired elements. You can use the placements of the (previous) transparent option (of the section) to help with the placement of the new devices.

As an editing aid, you can also copy the previous section before switching to transparent and place it over the now-enabled, transparent section. Now you can replace devices, add new devices, or delete unnecessary devices and other elements. Completely different wiring can also be created, depending on what partial circuit is desired or required for this option.

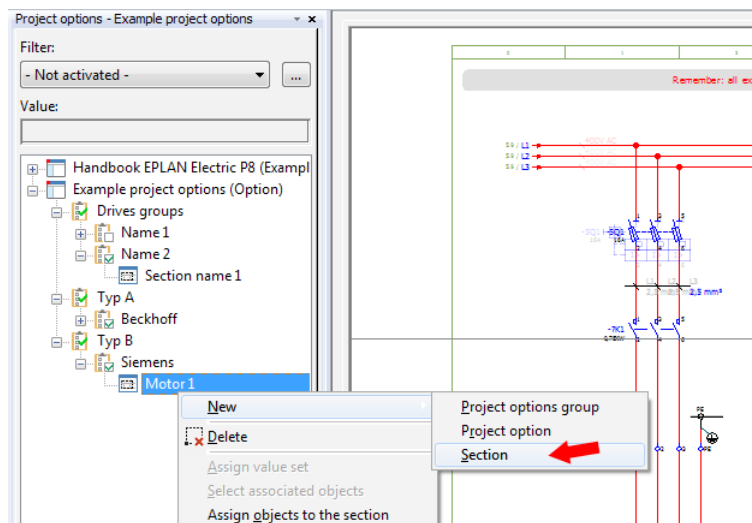


Fig. 12.35 Defining a new section

Once all changes for this section have been made, the section can be defined and saved. You again use windows to select the devices and save them with a click of the left mouse button. Then you can use the **PROJECT DATA/PROJECT OPTIONS/CREATE SECTION** menu or the popup window of the **project options navigator** to create the new section.

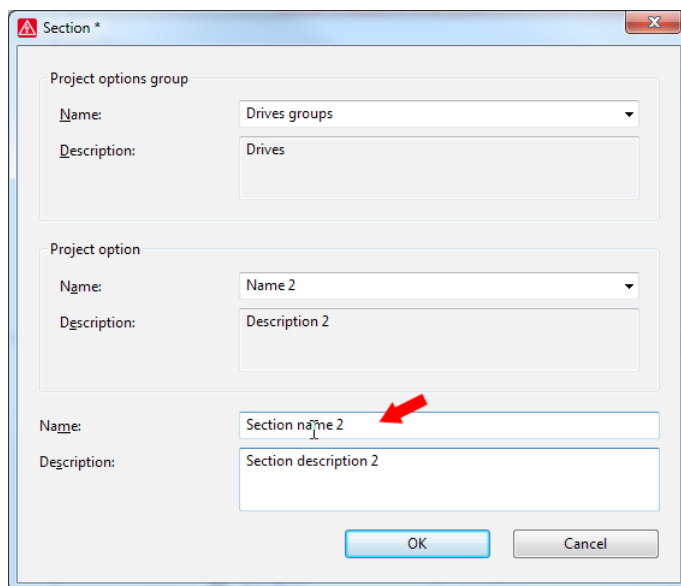


Fig. 12.36 Designating a new section

You give the new section a **name** and **description** as before. The section is then saved. Both sections are now available as options in the project and can be switched on and off as desired.

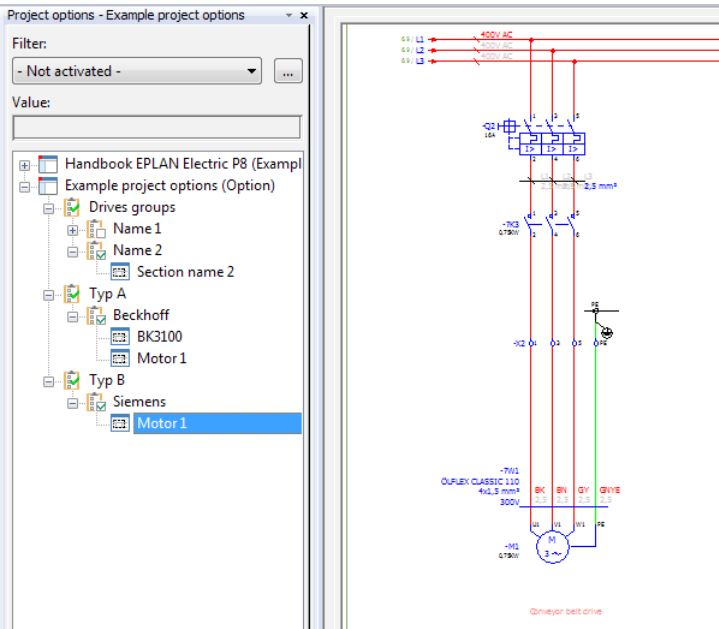


Fig. 12.37
Section with a motor

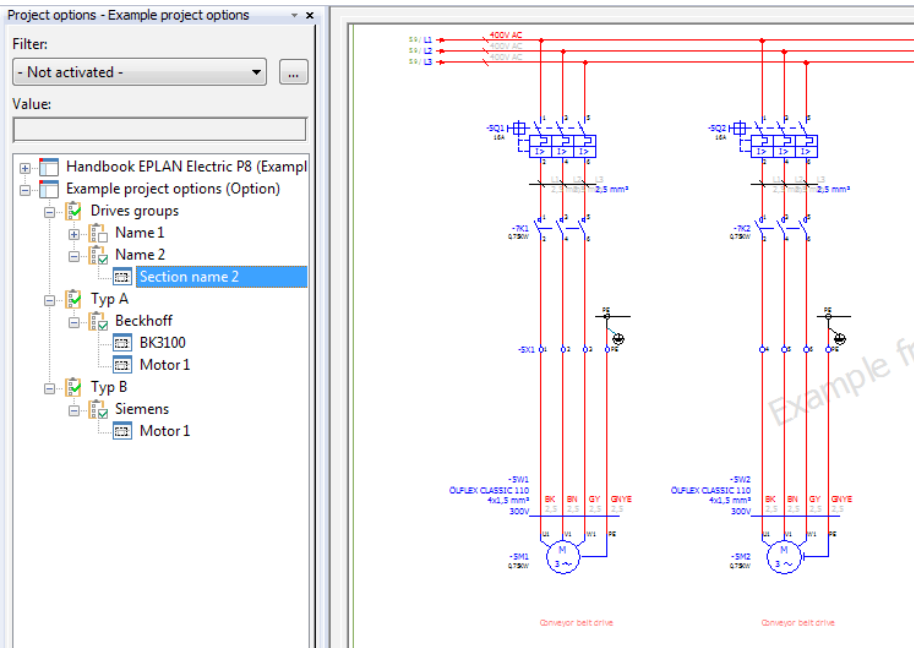


Fig. 12.38
The same page with
another section

12.2.4 Generate options overview report

If several options are being used in a project, it may be helpful to have an overview of the options that are used and not used. EPLAN offers an easy option for automatically creating reports in a broad range of forms. All that is required is a form of the **project options overview** type and a report or report template.

Fig. 12.39 shows an example of how this kind of overview of the options used could appear in a report. Of course, you can use filters and sortings here so that, for example, your report only includes the active options.

Project options - Example project options

Filter: - Not activated -

Value:

Handbook EPLAN Electric P8 (Example)
Example project options (Option)
Drives groups
Name 1
Name 2
Section name 2
Typ A
Beckhoff
BK3100
Motor 1
Typ B
Siemens
Motor 1

Remember: all examples are fictional examples and are only used for the purpose of these

Option overview

Option section name	Option section description	Option active	Option group name	Option group
Handbucher switch	Handbucher switch	x	Brückung	Power 10
Handbucher	Handbucher		Brückung	Power 10
Handbucher	Handbucher		Brückung	Power 10
Handbucher	Handbucher		Brückung	Power 10
SPS grüner Überwachen/SPS grüner Überwachen	SPS grüner Überwachen/SPS grüner Überwachen	x	SPS	PLC
SPS grüner Überwachen/SPS grüner Überwachen	SPS grüner Überwachen/SPS grüner Überwachen		SPS	PLC
1 Motor	1 Motor	x	Anteilung oben	Anteilung oben
2 Motor	2 Motor		Anteilung oben	Anteilung oben
			Anteilung oben	Anteilung oben
			Anteilung oben	Anteilung oben

Fig. 12.39 Automatically generated options overview

13

FAQs

Despite the extensive online help that EPLAN provides, there will always be one or the other question that is not covered or whose answer does not precisely fit the situation at hand. This chapter therefore contains a number of questions frequently asked about EPLAN Electric P8 and their answers, divided into categories.



NOTE: I should point out that some questions may have several answers. EPLAN Electric P8 is known for offering different solution approaches; there is not “just one right way”.

I would also like to point out the EPLAN Solution Center. You will find a series of FAQs there, as well as have the chance to ask your own questions that will then be answered directly by EPLAN experts (the EPLAN Solution Center requires registration).

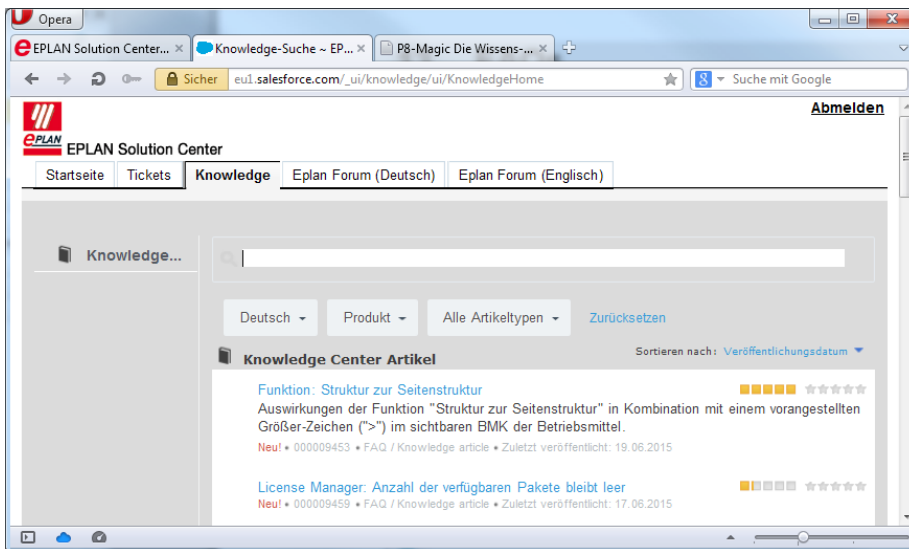


Fig. 13.1
EPLAN Solution Center
on the Internet

■ 13.1 General



Question: What is the fastest way to rotate grouped graphical elements?

Answer: Select the grouped elements, then define the center of rotation (EDIT/ROTATE menu) and use the OPTIONS/RELATIVE COORDINATE INPUT menu (or the SHIFT + R shortcut key) and the subsequent **Relative coordinate input** dialog to enter directly the desired angle of rotation for the grouped element. Confirm the selection with OK. EPLAN then rotates the element by the entered angle.

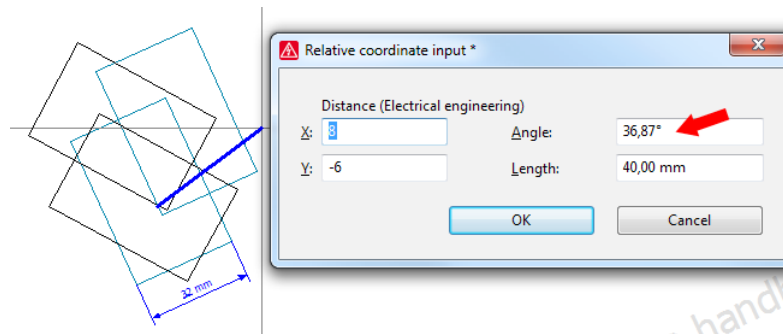


Fig. 13.2 Setting the angle in the dialog



Question: How do I enlarge or reduce graphical objects?

Answer: Select the object to be enlarged/reduced. In the EDIT/GRAPHIC menu, select the SCALE function. Specify the origin for scaling and confirm with OK.



NOTE: Only integer values or values with a decimal point are valid here. Negative values are not permitted.



Question: What is the meaning of the PROPERTIES (GLOBAL) function in the OPTIONS menu?

Answer: To edit properties globally, you have to activate the OPTIONS/PROPERTIES (GLOBAL) menu. This allows you to simultaneously (i.e. globally) edit an object (such as a contactor) and its functions (here: contacts) that are distributed throughout the schematic. For example, you could change the DT throughout the schematic for all functions in one go.

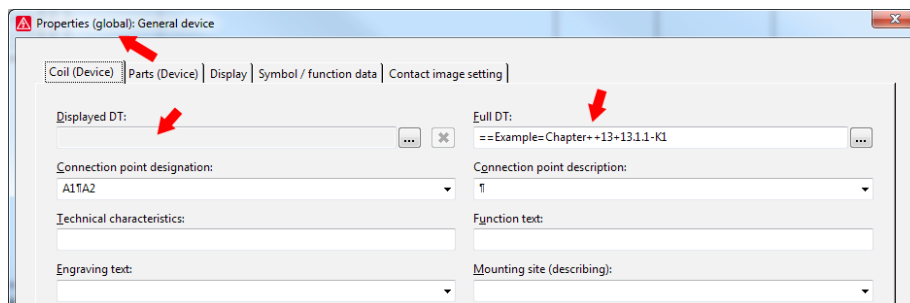


Fig. 13.3 Properties (global) option

When the **PROPERTIES (GLOBAL)** option has been activated, it is not possible to change the *Displayed DT* in the symbol properties. The field will appear grayed out. Only the *Full DT* field can be edited.

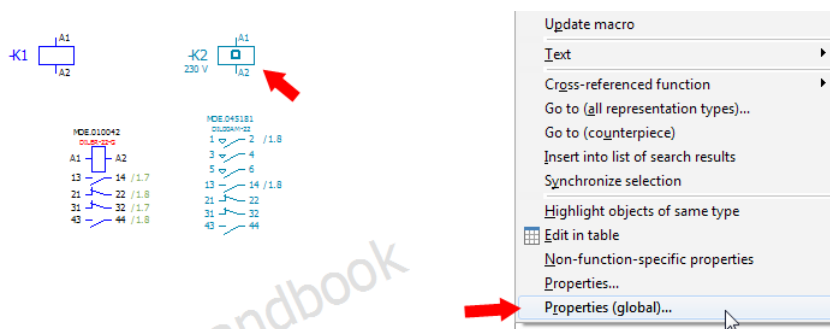


Fig. 13.4 Popup menu

The *Properties (global)* function can also be found in the right-click popup menu. To use this function, you must select, for example, a device. Then, you can call *Properties (global)* from the popup menu. Actually, you can select any device. It could be the coil, or any other function of the coil.

Question: What is the difference between a *basic project* and a *template project*?

Answer: *Project templates* always contain all project-specific settings and the schemes of the project structure. Project templates have the file extension *.ept. If a new project is created on the basis of a project template, all project-specific settings will be imported and applied. The page structure in such a newly created project can be adjusted subsequently if no pages have been added to the project yet.

Basic projects contain both pages and all project-specific settings. They have the file extension *.ebp. If a project is created on the basis of a basic project, all project-specific settings and the pages already contained in the basic project will be imported and applied. However, the page structure of the new project is defined by the pages of the basic project and, unlike project templates, cannot be changed.



The page numbering, though, can be changed for both types of projects. The main difference lies in how the page structure is modified. In one case it can be changed (new project from a project template), while in the other (new project from a basic project) it cannot be changed.



NOTE: As of version EPLAN Electric P8 1.9.x, basic projects always have the file extension **zw9*. The previous **epb* format still exists and can be used as a basis for new projects. However, it is no longer possible (this is also true of version 1.9.x and up) to create basic projects with the **epb* extension.



Question: How do I modify the line type of autoconnecting connections?

Answer: To modify the formatting of autoconnecting lines, you can set a *potential definition point* on the autoconnecting line and influence the formatting of the autoconnecting line in the symbol properties on the *Connection graphic* tab. A potential definition point can be inserted via the **INSERT/POTENTIAL DEFINITION POINT** menu.

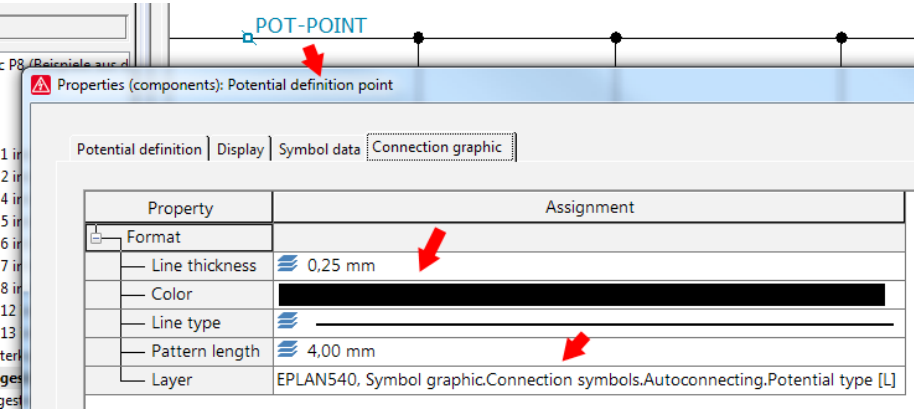


Fig. 13.5 Changing the formatting of autoconnecting lines globally

Since this involves a connection, the connections should then be updated (via the **PROJECT DATA/CONNECTIONS/UPDATE** menu). This way, all connections affected (e.g. PE connections) are updated, and they adopt the format of the potential definition point.

If you want to change the line type for a partial connection only, you must use a connection definition point. A connection definition point can be inserted via the **Insert/Connection definition point** menu, which can then be placed on the intended section of the connection.

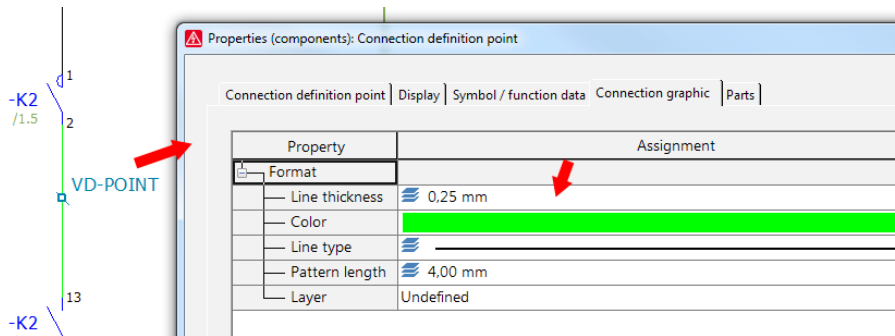


Fig. 13.6 Modifying the property of a partial connection

Question: Can macros with value sets be used across pages?

Answer: No. Macros with value sets can only be used on, or with, one page. If you wish to use macros with value sets across pages, you will have to use the **ProjectOptions** add-on (to be purchased separately).

Question: Why do my interruption points in the project always refer to themselves on each page?

Answer: To ensure that cross-references work for interruption points, you must observe the following.

- *First page:* Insert an outbound interruption point
- *Second page:* Insert an inbound and an outbound interruption point
- *Third page:* Insert an inbound and an outbound interruption point
- *Last page:* Insert an outbound interruption point

Interruption points are always paired. The head of an interruption point must always come together with the tail of the next interruption point; always head to tail. Now, the cross-references (if all other properties are set to default) will work, and the interruption points will no longer result in wrong references.

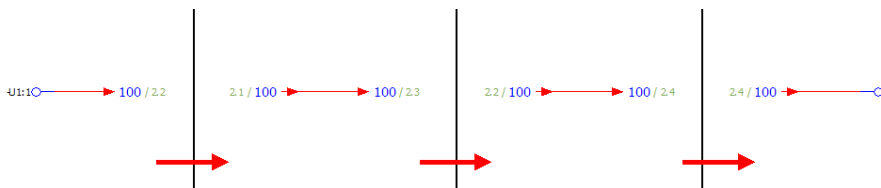


Fig. 13.7 Default sorting of interruption points

By defining a sort code for interruption points, it is also possible to connect specific interruption points with each other in a targeted manner (requires identical DTs). Again: There must always be a pair with the same sort code.

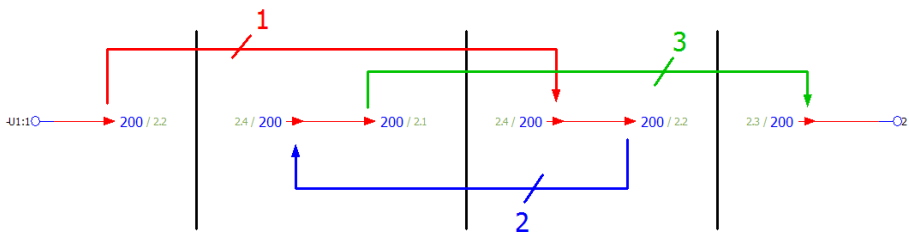


Fig. 13.8
Sorting interruption
points against default



Question: Why do I sometimes see red exclamation marks on my devices in the device navigator?

Answer: If you see a red exclamation mark on a device, it means that a message has been received for this device via the message navigator.

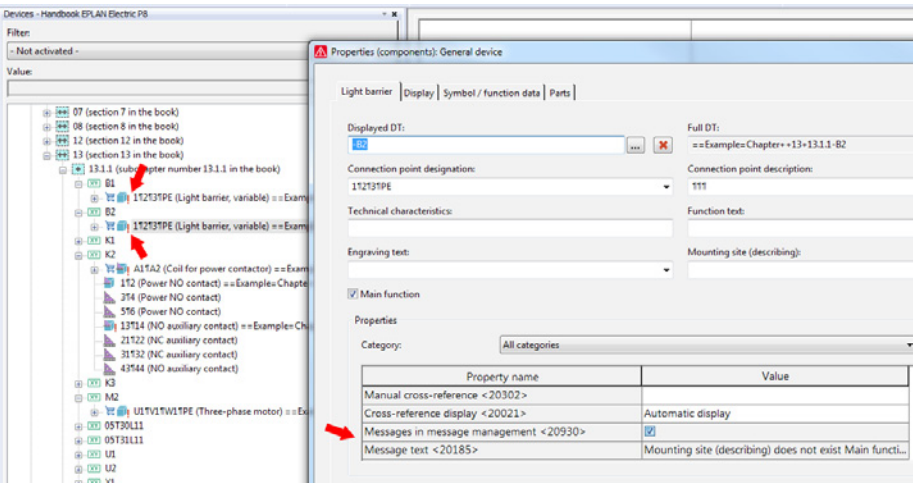


Fig. 13.9
Symbol properties

Simply open the message navigator, activate the *Selection* check box and select the device in the device navigator. The message or messages that have been generated for this device during the check run will be displayed.

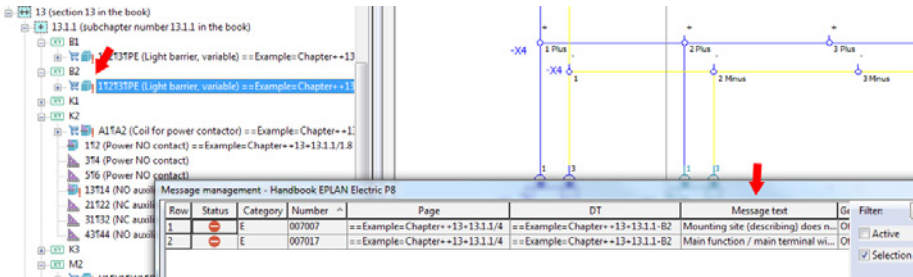


Fig. 13.10
Message in message
navigator

Question: Why does EPLAN Electric P8 not create connections, and why are no cross-references displayed?

Answer: In this case, the project in question is not of the **schematic project** type, but a project of the **macro project** type. These limitations are typical of macro projects.

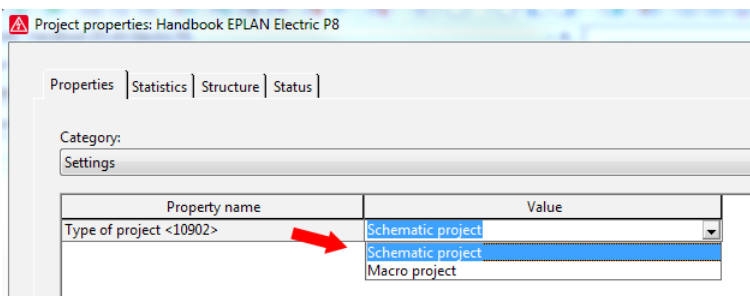


Fig. 13.11
Setting the project type

If this is not the type you want, you have to change the *Type of project <10902>* property in the project properties.

Question: How can I later change the project structure (page structure)?

Answer: You can't. The underlying structure of the project (page structure) cannot be later modified; under normal circumstances it can only be selected once (when creating the project).



NOTE: The page structure also cannot be changed when a new project is created from a basic project. This is also true if a new project is created based on a copy of another project. Here, too, it is not possible to change the page structure of the project, because it was defined as such in the project to be copied.

All other structures, such as devices, cables, etc. can, however, be later adjusted to the requirements of the project.



TIP: Generally, you can cut out all pages in the project in question (or generate a page macro for all pages). Then, as there are no more pages and thus no page structure in the project, you can modify the page structure in the project properties. Afterwards, you can insert the pages back into the project (or import the page macro that you just created).



NOTE: Any subsequent change to the page structure must be considered carefully, because this may also change fundamental settings for objects. This can produce unwanted results — up to the point that the project becomes useless. You should therefore always test things out on a copy of the project.



Question: Why doesn't copy and paste using CTRL + C and CTRL + V work anymore?

Answer: If copying with CTRL + C and inserting (pasting) with CTRL + V does not work (e.g. you select an object and then press CTRL + C to copy it and CTRL + V to paste it), then the DESIGN MODE in the OPTIONS menu has been activated. This should be disabled. The copy and paste functions will then again work as usual.



Question: What is the difference between *Delete* and *Delete placement*?

Answer:

Delete: The selected object (device) is deleted completely (graphical placement and any device information). It is no longer available in the project data or in any of the navigators.

Delete placement: The object is only “graphically” deleted. It is no longer visible graphically (e.g. on a schematic page), but the object itself still exists in the project data with all its device information, e.g. in the device navigator, and can be placed “graphically” again later on or be retrieved from the relevant navigator.



NOTE: The *Delete* function is also available in various navigators. If you select functions (unplaced and also placed objects) in the navigator and then delete them, they will be deleted definitively – including the placed ones. You should therefore be very careful when deleting things in navigators.

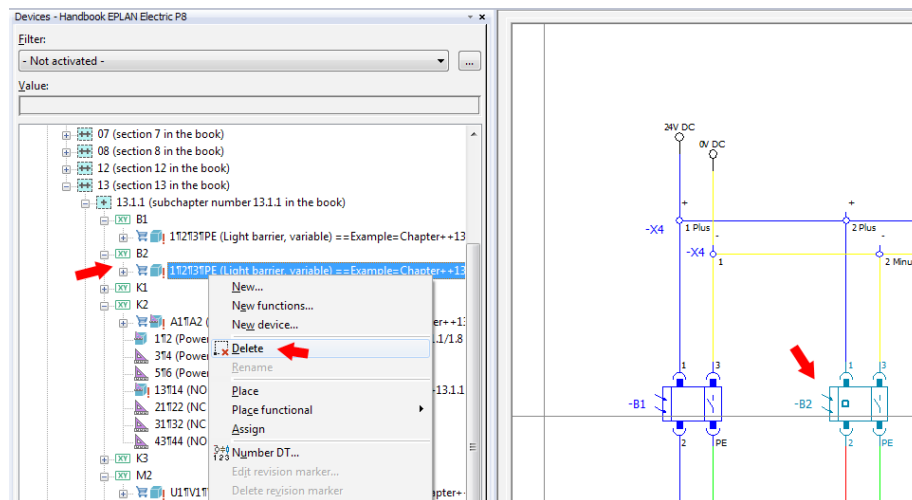


Fig. 13.12 Object selected to be deleted

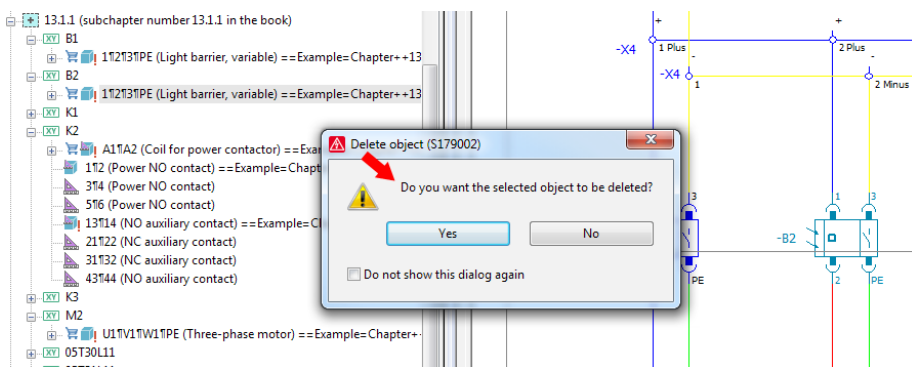


Fig. 13.13 Prompt before deleting the object

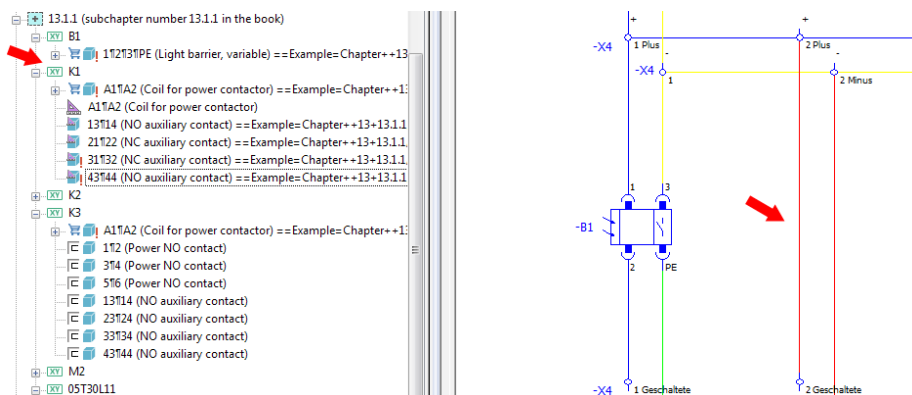


Fig. 13.14 The object has been deleted.

Question: Why can't I enter data into the "Displayed DT" field anymore (it is grayed out)?

Answer: If in the symbol properties the *Displayed DT* field is grayed out (i. e. no data can be entered here), then the *Properties (global)* option has been activated. This must first be deactivated to be able to edit the *Displayed DT* field as usual.

Question: Where can I find a demo version of EPLAN Electric P8?

Answer: There is a P8 version for students, with a trial license of 270 days. It is called EPLAN Education and can be downloaded from the EPLAN website. As well as EPLAN Electric P8, it also contains other EPLAN program packages, such as Fluid power or Pre-planning, depending on the stage of development. Installation instructions for the download are available.



NOTE: The data in the Education version is not compatible with the data in the industrial (purchased) version. And data from the industrial version cannot be edited with the Education version.





Question: How can I edit a single object within a grouping?

Answer: It is possible to edit a single object embedded in a grouping (without having to undo the grouping) by keeping the **SHIFT** key pressed and double-clicking the intended element. EPLAN then opens exactly this one object, and you can modify its data.



Question: How can I reduce five lines simultaneously by a specific length in EPLAN?

Answer: By using the stretch function. This function stems from the CAD area and works as follows (ideally, the **OBJECT SNAP** mode should be activated in the **OPTIONS** menu). Call the **STRETCH** function via the **EDIT/GRAPHIC** menu.

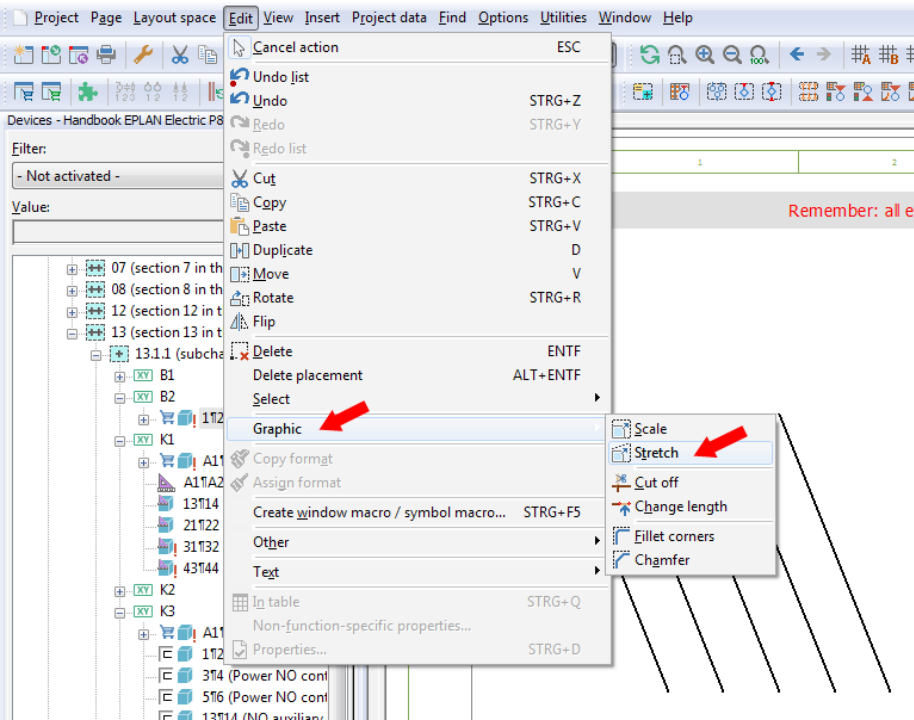


Fig. 13.15
Stretch function

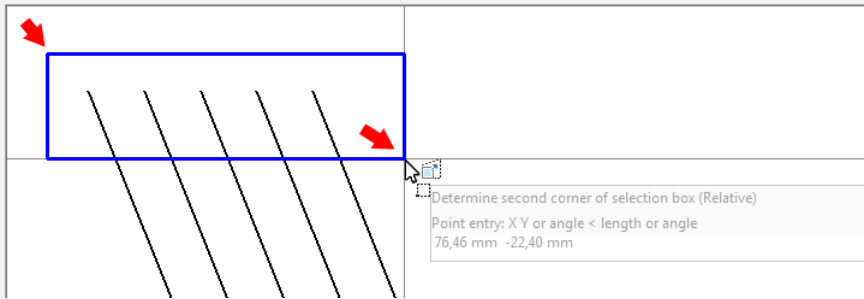


Fig. 13.16
Pulling open a window

Now, you select all line ends by pulling open a window (left mouse button) (the ends are “framed”). EPLAN marks the ends of the lines with a circle.

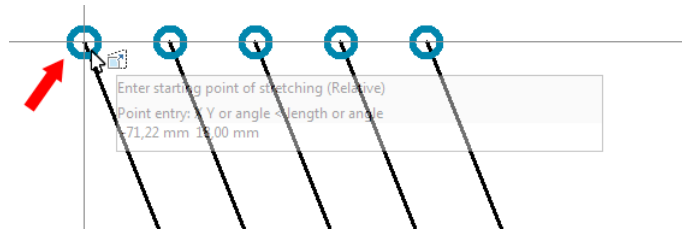


Fig. 13.17 Defining the starting point of the stretch

Then define the starting point of the stretch. Usually, this is done by clicking the end of a line. If object snap is enabled, the cursor will “snap” into place (red rectangle).

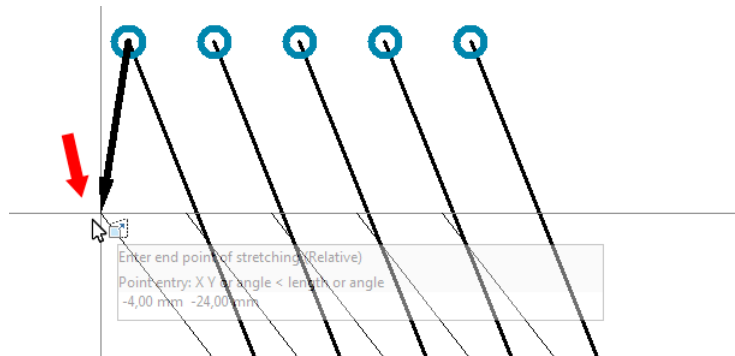


Fig. 13.18 Defining the end point of the stretch

Then you define the end point of the stretch (can be selected via the left mouse button; until it is placed EPLAN displays a cursor with a blue arrow). Any end point can be selected.

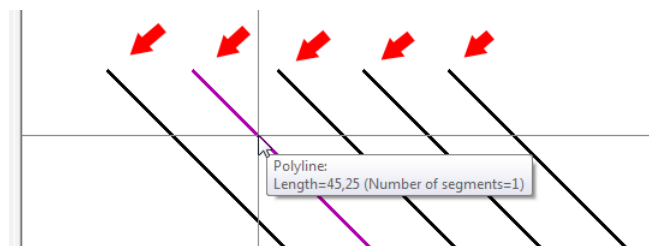


Fig. 13.19 All line data has been changed.

When you have selected the placement end point, confirm it via the left mouse button (click once). Now, the lines are “stretched” until this end point, i.e. shortened in this case. Of course, it also possible to extend them or to change the angle for all of them.



Question: How can the creator, customer or end customer data in the project properties be filled in automatically?

Answer: To avoid having to always manually enter the creator, customer or end customer data, such as address, name, etc. in the project properties, EPLAN provides an option that allows this data (which, of course, must already have been set up in parts management in the **Customer** area) to be written into the project properties by EPLAN automatically.

Open the (optional) *Project management* in the **PROJECT** menu and select the intended project. Now click the **EXTRAS** button and then select the **READ CUSTOMER DATA** function. Here, you can select the data for the creator, customer or end customer from the **Select address** dialog. Click the **OK** button to import them into the project properties. EPLAN then fills automatically the corresponding project properties with the data stored in parts management.



NOTE: Without optional project management, the customer data must be entered in the project properties manually, unfortunately.



Question: What is the difference between *Project properties* and *Project settings*?

Answer: *Project properties* are properties of the project, such as settings for the project structure or the types of devices used in it, such as terminal strips or general devices. Also part of project properties are specific properties that apply to the entire project globally, such as properties of the creator or the date of the project start. Project properties also include statistical information, such as the number of specific page types, etc. Project properties are called via the **PROJECT** menu and the **PROPERTIES** menu item. Alternatively you can select the project in the **page navigator**, then open the popup menu and select **PROJECT** and then **PROPERTIES**.

Project settings are settings that affect the project but have nothing to do with the project “content”, such as which forms are to be used for project reports, which online DT numbering scheme should be used or how the project structure should be displayed in the page navigator. You use the **OPTIONS/SETTINGS** menu to call up the project settings. Then, the correct project must be selected in the **Projects** node (in case of several open projects).



Question: How can I interrupt an Autoconnecting line?

Answer: An Autoconnecting line can be interrupted by means of the ACBP symbol (Auto-connect break point). The symbol can be inserted by pressing **CTRL + SHIFT + U**, or can be selected in the **INSERT/CONNECTION SYMBOL/BREAK POINT** menu and then placed randomly like any other symbol.



Question: How can I change the position of the contact image?

Answer: The position of the contact image can be modified in different ways, e.g. in the page properties of the current page by changing the *Contact image offset <12061>* property. This value applies only to the current page and all symbols affected by it.

The position can be changed globally by adjusting the plot frame in use. To do this, adjust the *Contact image offset* <12061> property in the plot frame editor. This setting will then apply to the entire project.

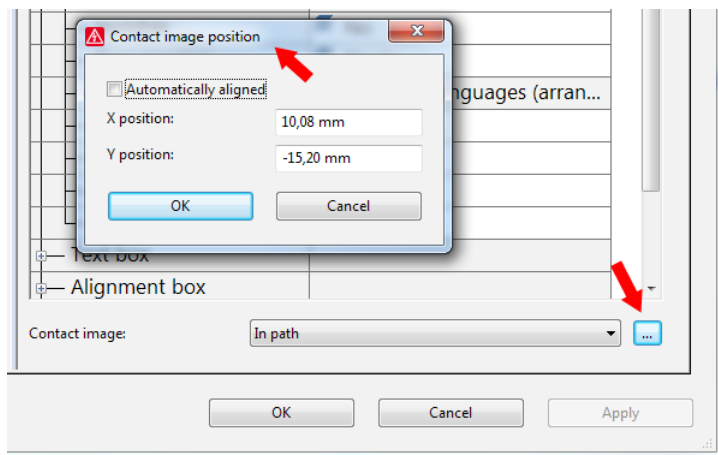


Fig. 13.20 Manually changing the position of the contact image

A third option involves calling the **Contact image position** dialog by using the **MORE** button directly on the affected symbol on the *Display* tab in the *Contact image* area and manually defining the X and Y positions.

It is also possible to select the symbol, press the **CTRL + B** shortcut key and click the contact image, while moving it into any position while keeping the left mouse button pressed.

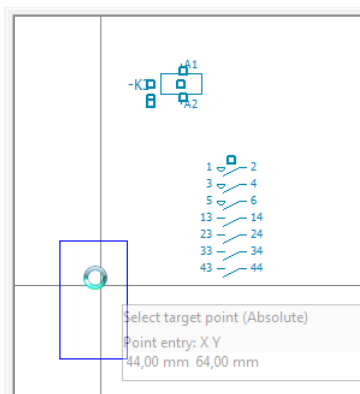


Fig. 13.21 Moving

Question: How do I find the previous “Superior” option in the project structure in case of different devices?

Answer: If the *Superior* option is no longer part of the project properties (*Structure* tab), you can reactivate it in the project's settings. You find the setting in the node **PROJECTS/[PROJECT NAME]/MANAGEMENT/COMPATIBILITY**.

But you should do without it, for reasons concerning norm-compliant representation. It is automatically activated in projects that have used this setting so far. But in new projects it is deactivated by default.



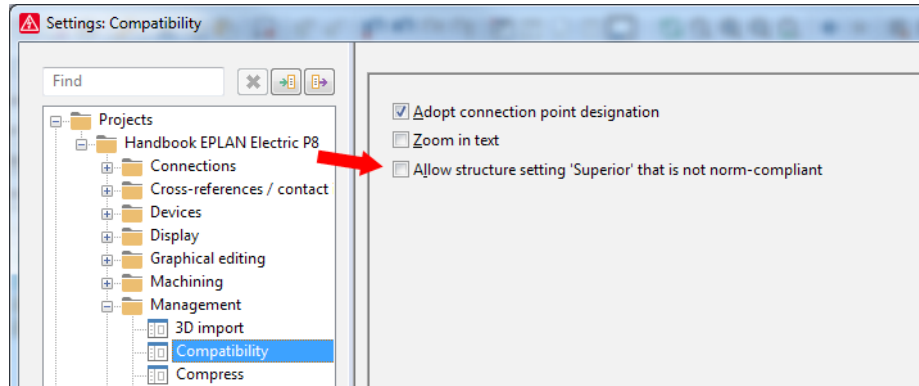


Fig. 13.22 Activating “Superior...” project setting

The setting does the following: If the device has the preceding sign “-”, the device will be allocated the structure identifiers (of the page, etc.) as defined in the project structure. If the devices does not have the preceding sign “-”, this does not happen.

For example, you can take one terminal strip and have it created globally across several mounting locations by omitting the “-” preceding sign. At the same time, it is also possible to generate a report on the remaining terminal strips, depending on the project structure defined, based on higher-level function and mounting location by placing the “-” preceding sign.

The *Superior* option is possible for the following device types: terminal strips, plugs, cables and interruption points.



Question: Where and how can I re-sort my structure identifiers?

Answer: To re-sort structure identifiers such as higher-level function or mounting location, you use the structure identifier management tool. It can be found in the **PROJECT DATA/STRUCTURE IDENTIFIER MANAGEMENT** menu. Here you will find editing options to re-sort or rename identifiers, to create new identifiers or to edit existing or create new descriptions of structure identifiers, and much more.



Question: How can I display the message(s) in message management for a specific object?

Answer: To view the relevant messages from message management for an item in the device navigator, you need to take the following steps.

1. Launch **PROJECT DATA/MESSAGES/MANAGEMENT** message management with the proper check run.
2. In the message management window, activate the check box next to **SELECTION**.
3. Start the device navigator.
4. Click the desired (defective) item in the device navigator.
5. Now you will see only messages in message management that concern this item.

Question: Where in EPLAN can I enter my own connection point designations or connection point descriptions for symbols?

Answer: For example, if you want to have for a horn (signal device) the existing connection point designations [x1 x2] as well as the connection point designation [L N] (in the symbol properties dialog of the horn), you need to take the following steps:



NOTE: Since this involves original EPLAN data records, these data records are reset to the EPLAN default in case of any version change. Keep this in mind when making changes.

1. Via the **UTILITIES/MASTER DATA** menu, open the **CONNECTION POINT DESIGNATION/DESCRIPTION** menu item.
2. On the *Connection point designations*¹ tab, locate the corresponding function definition (here: *Signal device, acoustic, single*) in the *function definition* column.

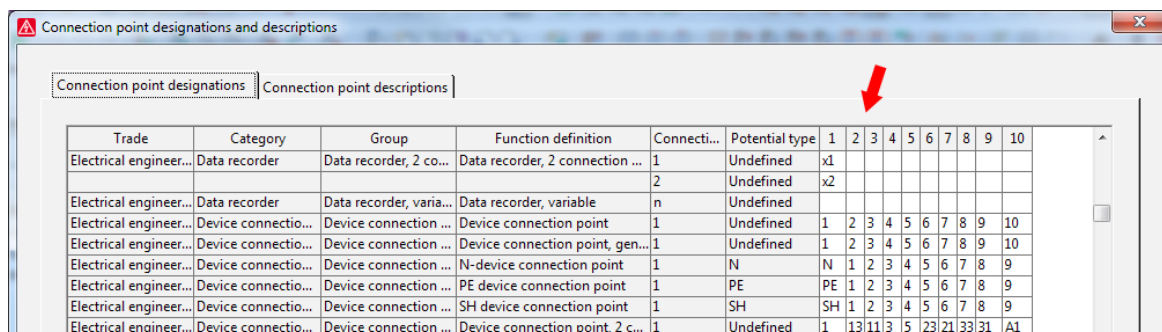


Fig. 13.23 Adding connection point designations

3. In column 2, enter the additional connection point designations per connection point.
4. Click the **APPLY** button and close the dialog with **OK**.
5. In the **UTILITIES/MASTER DATA** menu, select the **UPDATE CURRENT PROJECT** menu item. EPLAN now updates the project master data.
6. The new connection point designation [L N] is immediately available at the horn for selection.

Question: What is the difference between *[Update current project]* and *[Synchronize current project]*?

Answer:

Update current project: EPLAN automatically updates the project master data with the system master data. EPLAN will try, for instance, to store master data that is used in the

¹ For your own connection point descriptions, follow the same procedure. The only difference is that you must select the *Connection point descriptions* tab in the dialog.

project, but was not found, or not in full. Manual editing is not possible at this point. Once this function is launched, there will be no more confirmation prompt (to abort the function, you would have to click the **CANCEL** button in the dialog). After the update, EPLAN will display a message.

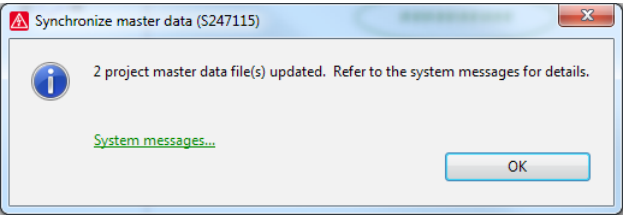


Fig. 13.24
Message regarding synchronization of master data

Synchronize current project: First you open the **Synchronize master data** dialog. In this dialog, master data can be exchanged in a targeted manner (in both directions, i.e. system to project or project to system). It is therefore possible to manually update specific master data (synchronize).

Both menu items are found in the **UTILITIES/MASTER DATA** menu.



Question: Can EPLAN create a macro from several pages?

Answer: Yes, EPLAN can do that. Such macros are known as page macros in EPLAN.

A page macro of a page can be created via the **CTRL+ F10** shortcut key. Alternatively, you can select the page in the page navigator and then select in the popup menu (right mouse button) the **CREATE PAGE MACRO** item. Then the information has to be stored in the **Save as** dialog.

The procedure for creating a page macro for multiple pages is similar. You select the pages in the page navigator and then press **CTRL + F10** or right click to call up the popup menu and call up the **Create page macro** menu item.



Question: Why are blue boxes instead of graphical symbols shown in the reports?

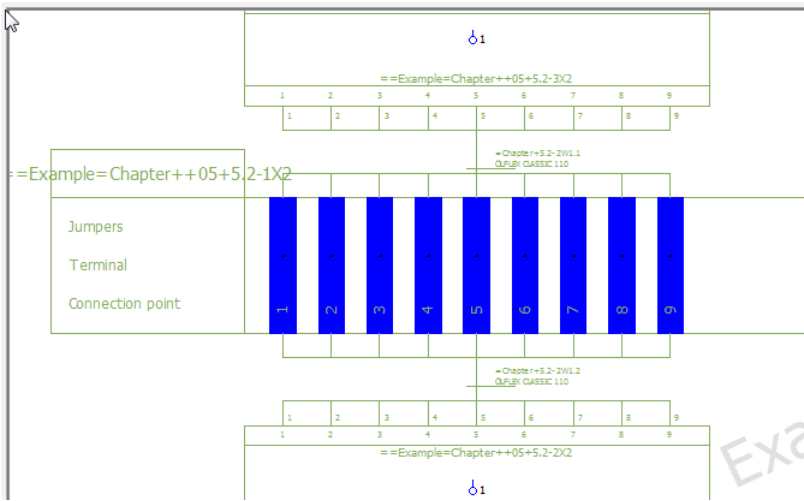


Fig. 13.25
Blue boxes

Answer: For graphical symbols, for example, from the GRAPHICS symbol library, to be displayed in reports, such as terminal or cable diagrams, the following user setting must be enabled. Under **OPTIONS/SETTINGS/USER/GRAPHICAL EDITING/2D** and here, in the selection field for the *Graphical symbol* color setting, you must select the *From symbol* setting.

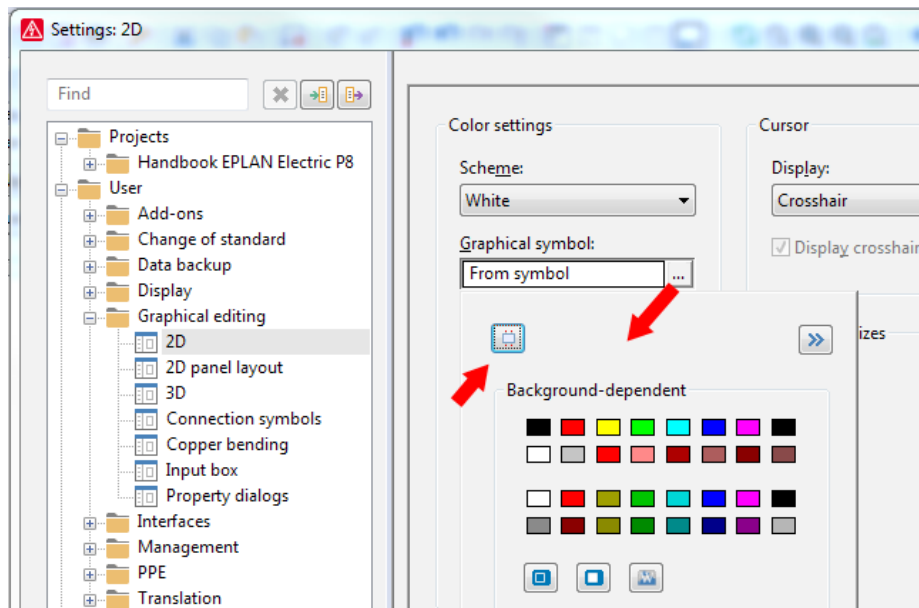


Fig. 13.26
Setting the color from the symbol

Now the graphical symbols will be displayed completely even in the reports.

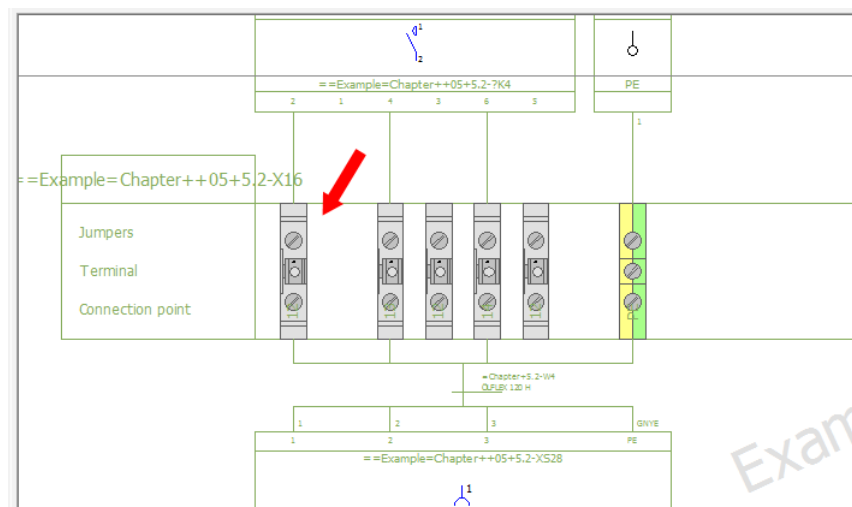


Fig. 13.27
Symbol color settings



Question: What is the difference between a window macro and a symbol macro?

Answer: Generally, they are not vastly different. Both macro types (window *.ema and symbol macro *.ems) can combine and store parts of a page or different objects in a macro, which can then be retrieved as a window or symbol macro.

A window macro is called via the **M** key or via the **INSERT/WINDOW MACRO** menu; a symbol macro via the **CTRL + INS** shortcut key or via the **INSERT/SYMBOL MACRO** menu. Both can be placed anywhere on the page.



Question: What is a main function?

Answer: A *main function* is the “leading” device with (distributed) devices displayed (e.g. the coil of a contactor) or a single switch. Only the main function, e.g. bears the device definition (parts data) of the device. Automatically, all other possible functions (such as the coil contacts) are not main functions or auxiliary functions.

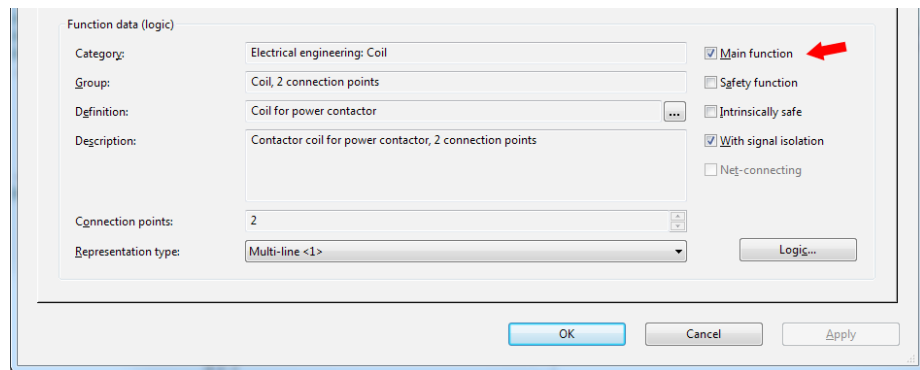


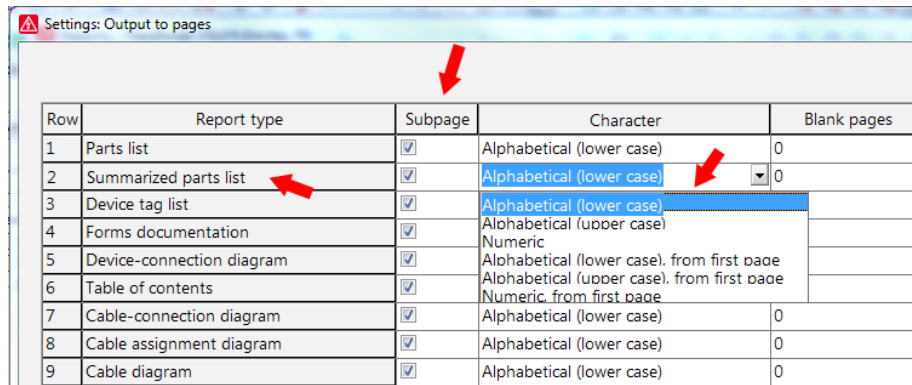
Fig. 13.28 Coil main function

In EPLAN, each device used can only have exactly *one* main function. This is true of all objects, whether they be contactors, motor overload switches or cable definitions.



Question: Where are the page sortings (subpage, characters) of reports stored?

Answer: To set up the desired sorting of report pages (subpages yes/no, numeric, letters uppercase/lowercase), go to the **UTILITIES/REPORTS/GENERATE** menu and open the **Reports** dialog. Here, click the **SETTINGS** button and then select the **OUTPUT TO PAGES** menu item. In the dialog, set the desired *Sorting* in the *Character* column and whether subpages (*subpage* column) are to be created. Finally close and exit the dialog with **OK**. Now the reports can be generated and updated.



Row	Report type	Subpage	Character	Blank pages
1	Parts list	☒	Alphabetical (lower case)	0
2	Summarized parts list	☒	Alphabetical (lower case)	0
3	Device tag list	☒	Alphabetical (lower case)	
4	Forms documentation	☒	Alphabetical (upper case)	
5	Device-connection diagram	☒	Numeric	
6	Table of contents	☒	Alphabetical (lower case), from first page	
7	Cable-connection diagram	☒	Alphabetical (upper case), from first page	
8	Cable assignment diagram	☒	Numeric, from first page	
9	Cable diagram	☒	Alphabetical (lower case)	0

Fig. 13.29 Settings: Output to pages dialog

Question: What project information appears in the **Open project** dialog on the right-hand side of the dialog?

Answer: The project information in the **Open project** dialog (on the right-hand side in the preview window) are taken from the project properties. This information cannot be affected or replaced with other project properties.

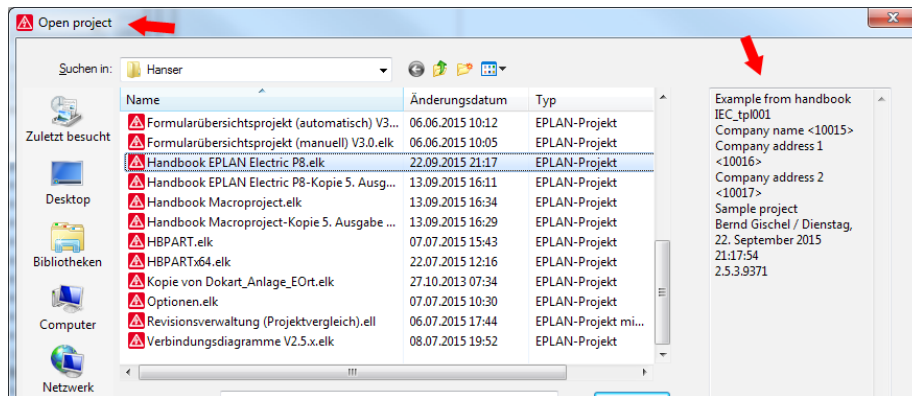


Fig. 13.30 Information in the Open project dialog

The following project properties are displayed (if they have been completed in the project properties):

Project description <10011>;

Project number <10013>;

Commission <10014>;

Company name <10015>;

Company address 1 <10016>;

Company address 2 <10017>;

Supplementary field [1] <10901 1> to Supplementary field [7] <10901 7>;

Last edited by: Logon name <10022>;
 Modification date <10023>;
 Last EPLAN version used <10043>;
 Last EPLAN build number used <10044>



Question: How can I restore EPLAN to the values of the default installation?

Answer: To restore the default installation values of EPLAN (when confusing situations arise that cannot be explained or removed in any other way), EPLAN can be started by calling up: “C:\Program Files\EPLAN Electric P8\2.5\BIN\Eplan.exe” /setup, where the program directory shown here must be adjusted to reflect your own installation directory.



NOTE: When you launch this call-up, you will lose all personal settings, such as set schemes, workspace, etc. These settings must be saved beforehand.



Question: The object snap for the dimension function does not work anymore.

Answer: If graphical objects on pages – e.g. for the dimension function – are no longer automatically captured by the mouse (the small red rectangle), then the **OBJECT SNAP** in the **OPTIONS** menu has been deactivated.

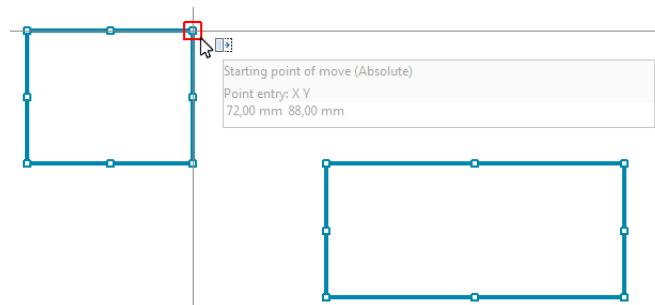


Fig. 13.31
Object snap on

Once object snap is activated, you will be able to capture objects (points) again.



Question: How can I create completely new connections or repair connections?

Answer: If connections (or their additional information) exhibit strange behavior, it is possible to recreate all project connections from scratch. To do so, keep the **SHIFT** and **CTRL** keys pressed and call from the **PROJECT DATA/ CONNECTIONS** menu the **UPDATE** menu item. Now, all connections (and/or the connection database) will be recreated. Depending on the number of connections in the project, this may take some time, because EPLAN first deletes all connection data and then recreates them fully from scratch.



Question: How can I later change the sorting of the structure identifiers, such as higher-level function or mounting location?

Answer: To sort identifiers (structure identifiers), such as higher-level function or mounting location, you must start **STRUCTURE IDENTIFIER MANAGEMENT** from the **PROJECT DATA** menu. In the dialog that follows, you can change the sequence of the various iden-

tifiers used in the project (and also those not currently in use). The identifier is selected from the identifier list in the left area of the **Structure identifier management** dialog.

Question: Can EPLAN also process page names like “Seventy”?

Answer: Yes, this is possible. Usually the page name consists of a number (page number). For example, the name of the page can be “70”. EPLAN can also include alphanumeric components in the page numbers. Accordingly, a page can also be called “Seventy”. However, any subsequent operations such as numbering pages will no longer function as desired since the pages have alphanumeric names.



The “number” property of the page is called page “name” and not “number” because it can be alphanumeric.

Question: How is it possible, for example, that a device connection point with the potential type PE has the same designation, without EPLAN showing a project check message?

Answer: In this case, the **LOGIC** (on the *Symbol/function data* tab and, here, the **LOGIC** button) of the device connection point must be adjusted. In the following **Connection point logic** dialog, there is the *Allow same connection point designations* setting. This setting must be activated for the relevant device connection points. This will keep EPLAN from objecting to functions having identical designations.

Question: Does the sequence of the stored symbol libraries affect the representation of the symbols?

Answer: No. The sequence of the stored symbol libraries does not affect EPLAN projects. EPLAN always takes symbols only from the correct symbol library. It relies on information such as the name of the symbol library, the symbol number and the symbol name. This requires, of course, that the same symbol libraries are/have been stored in the various projects.

Question: Why, after inserting a macro, do I see a bunch of red crosses instead of the symbol graphics?

Answer: After inserting a macro or similar things, if you see a bunch of red crosses instead of symbol graphics, the cause is often a missing symbol library in the project.

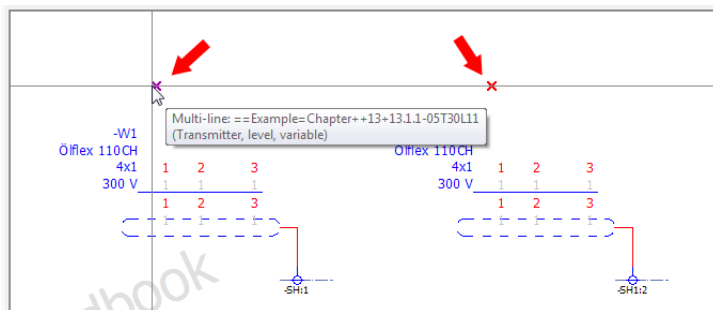


Fig. 13.32
Missing symbols

Since the symbol library/libraries is/are not stored (in the macro, etc.) when creating a macro or similar things (in another project), EPLAN needs the information to display the proper symbol graphic contained in the symbol library that was used in the project (where the macro was created).

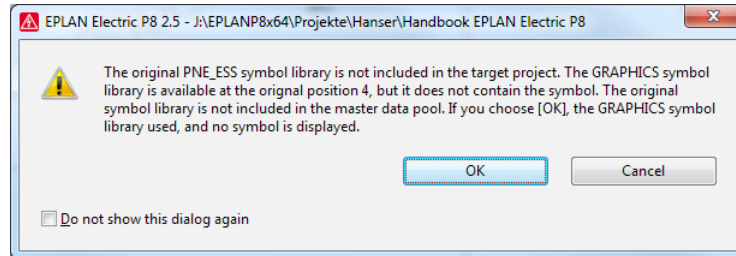


Fig. 13.33 EPLAN warning message when symbols are missing

This can be fixed by placing/storing the corresponding (missing) symbol library in the current project. Then, you must re-insert the macro, or simply select the symbol again.



Question: How can I insert special characters in EPLAN that do not exist in the **Special characters** dialog?

Answer: To insert special characters in texts or similar things in EPLAN, use the CTRL + S shortcut key to call the **Special characters** dialog. But the selection offered there is limited. If you want to insert a special character that is not included in EPLAN by default, you can copy it from another Windows application, for example, WORD, and then paste it into EPLAN Electric P8.



Question: A project was deleted by mistake. What can I do now?

Answer: If EPLAN was used to delete a project from the server, it is irretrievably lost, because Windows, the operating system, does not have a recycle bin (trash can) on the server. If the project was deleted from the local hard disk, you can restore the project from the Windows recycle bin.



NOTE: To restore projects deleted from the server, the only option is the company data backup you (hopefully) created.



Question: Can I change the automatic distance of the contact image on component?

Answer: Yes. To change the automatic distance of the contact image on a component, you must call up the *Symbol properties*. Then switch to the *Display* tab, and in the *Contact image* area, deactivate the automation and enter a new, manual value for the X/Y position (offset).



Question: How do I access the EPLAN Data Portal?

Answer: Anyone who has a software maintenance contract will receive immediate (free) access to the Portal and the data contained therein after creating a user account.

Question: What is the easiest way to automatically rename a structure identifier, for example, a mounting location?



Answer: The easiest way to automatically globally rename a *structure identifier*, such as a mounting location +OT12 to +U23, i.e. across the entire project, is to use *structure identifier management*. The mounting location used in the project, +OT12, will be found anywhere and renamed to the new structure identifier mounting location, +U23, without having to make manual changes. The structure identifier management tool can be accessed from the **PROJECT DATA** menu followed by the *Structure identifier management* menu item.

Question: In the **Select scope of menu** start dialog, what do the terms Beginner, Advanced and Expert mean?



Answer: The three settings – *Beginner*, *Advanced* and *Expert* - correspond to three fixed and defined user groups that are selectable if EPLAN is run without rights management.

Beginner refers to users with access to the functionalities that are absolutely necessary to draw schematics as well as work with macros and project data dialogs.

Advanced users can use other functions as well the previous ones (e.g. display options such as minimum text size, display of empty text boxes, etc.). They can also use data transfer.

Expert users have access to all functionalities, i.e. they can use the functions they need to prepare work such as system settings, edit master data, work with revisions and project options, back up data, etc.



NOTE: Conversely, this means: If one or the other function is missing, but according to the license should exist, check again to see the scope of menu with which EPLAN was launched.

Question: How and where can I change the view of the workspace?



Answer: Via the **VIEW/WORKSPACE** menu, it is possible to select from default views. In addition, you can create and save new workspace views of your own. As well, it is possible to save anew (overwrite) the current workspace with all its settings to a current scheme.

Question: How do I activate the mode for the direct editing of property texts?



Answer: In EPLAN, you can edit certain properties directly without having to call up the symbol properties dialog.

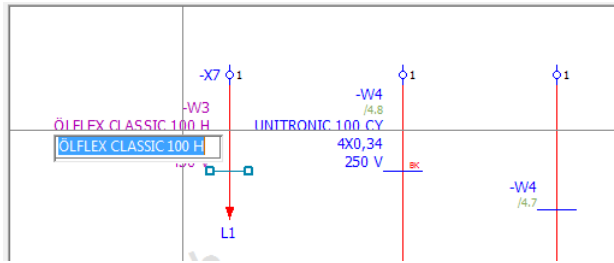


Fig. 13.34 Direct editing of properties

Apart from the option of enabling and disabling this edit mode via the **OPTIONS** menu, you can also activate direct editing temporarily using the mouse or keyboard.

Using the mouse: As soon as you have selected text or a component, keep the left mouse button pressed for a short amount of time. A small window opens where you can edit the data directly.

Using the keyboard: Move the cursor near the text or component and press the [F2] key. A small window again opens for direct editing.



TIP: In the case of several properties on symbols, after activating the direct input mode, you can use Tab to “jump” to the individual, (mostly) filled properties and edit them.



Question: How and where can I change the color scheme (background color) of the graphical editor?

Answer: Under **OPTIONS/SETTINGS/USER/GRAPHICAL EDITING/2D**, the color scheme of the graphical editor can be changed. The names of the color schemes indicate the color of the screen background: So, you can choose a *white*, *black* or *gray* background. The colors of the various elements are adjusted for the respective background. I recommend using the white background.

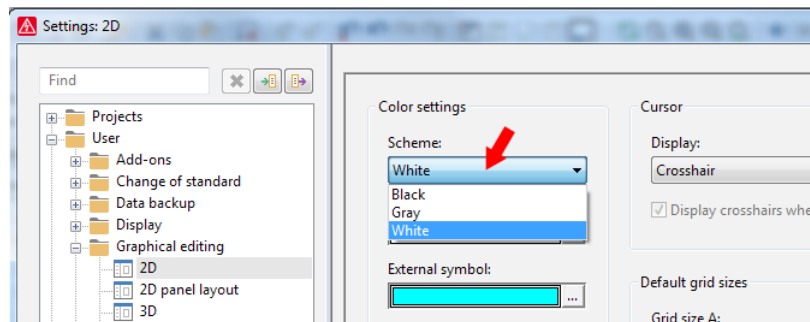


Fig. 13.35 Setting the background color of the graphical editor

Question: How can I re-activate suppressed dialogs/messages?

Answer: To reactivate dialogs that have been deactivated before (e.g. Tip of the day), you must again set the check mark in the user settings. Go to the **OPTIONS/SETTINGS** menu and select in the now open dialog **USER/DISPLAY/USER INTERFACE** from the tree structure. There, on the right-hand side, you can select the *Reactivate suppressed messages* option. Then close the dialog again. All deactivated messages/dialogs are available again immediately. For some dialogs/messages, it may be necessary to restart EPLAN.



Question: How do I save a project or a page in EPLAN?

Answer: It is not at all necessary to save a project or page in EPLAN. Every change is directly “written” = saved (on the storage medium, such as a hard disk). As a result, additional manual saving is not necessary.



Question: How can the sequence of projects in the page navigator be changed?

Answer: It cannot be changed. The sequence of projects (in the **page navigator**) depends on the sequence in which they were opened. The sequence of projects (position in the navigator) cannot be changed manually.



Question: Can I hide the description of structure identifiers in the page navigator?

Answer: Yes. This and other structural descriptions can be displayed and/or hidden via the popup menu using the *Configure representation* function.

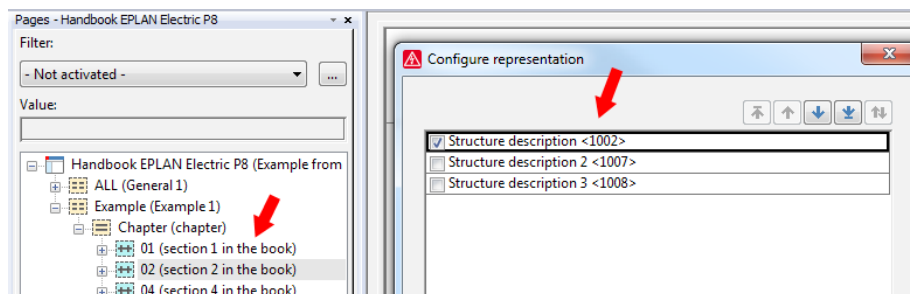


Fig. 13.36 Displaying and/or hiding structural descriptions in the page navigator

Question: Is it possible to install EPLAN Electric P8 (all versions) side by side with a previous version, such as EPLAN 5 version (also all versions)?

Answer: Yes, absolutely, because both programs operate separately from each other.



Question: Is it possible to increase the maximum number of entries (12) in the clipboard?

Answer: No. The clipboard is limited to 12 entries. The clipboard >1 is activated as follows: Go to the **OPTIONS/SETTINGS** menu, open the **USER/DISPLAY/GENERAL** node and adjust the clipboard settings in the right area of the dialog.



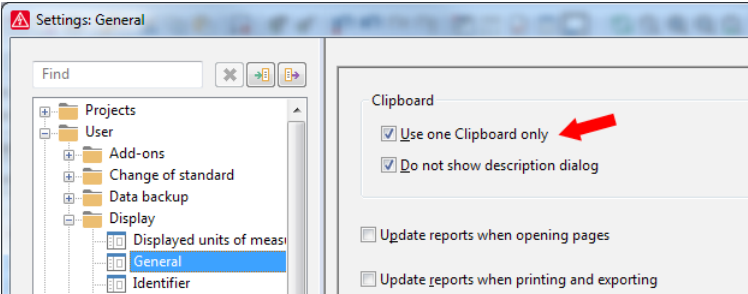


Fig. 13.37 Clipboard settings

 **NOTE:** If the maximum clipboard limit has been reached, EPLAN will display a warning.

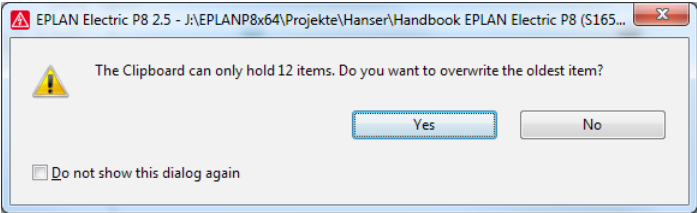


Fig. 13.38 Warning: Maximum clipboard limit exceeded.



Question: Can the *Update connections* function be assigned to a shortcut key?

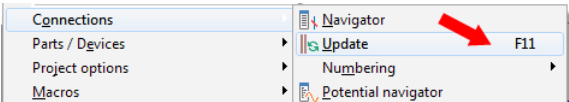


Fig. 13.39 User's own shortcut key F11

Answer: Yes, this is possible. In fact, it is recommended, because this function is used quite frequently. I recommend that you use the F11 function key. EPLAN currently does not use this key, and it is also easy to reach and remember.

You can create your own shortcut keys using the **OPTIONS/SETTINGS/USER/MANAGEMENT/SHORTCUT KEYS** menu. Then, you select the respective command and assign to it a suitable shortcut key.



Question: How can I suppress the check message “Coil without contacts”?

Answer: If the project uses contactors/relays, etc. as spares (without using contacts or parts), the check message “P007009: Coil without contact” might be displayed.

Solution 1: This can be remedied only if you assign this coil a part with a correct function template, i.e. once you have made a device selection. Then, all spare coils in the project will be ignored for the purposes of the check run messages.

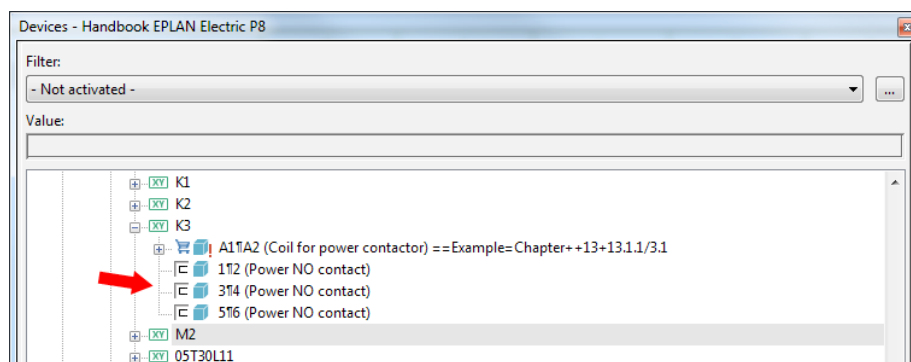


Fig. 13.40 Unplaced functions

Solution 2: If parts cannot be assigned, the missing functions, for example, can be generated automatically in the device navigator via the popup menu and the *New function* function.

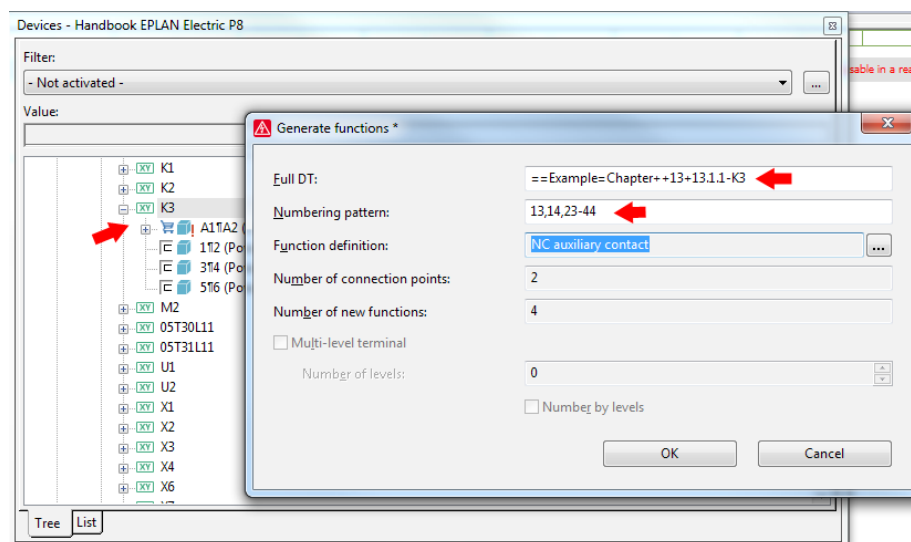


Fig. 13.41 Generating new functions with numbering pattern (here: 4 close contacts).

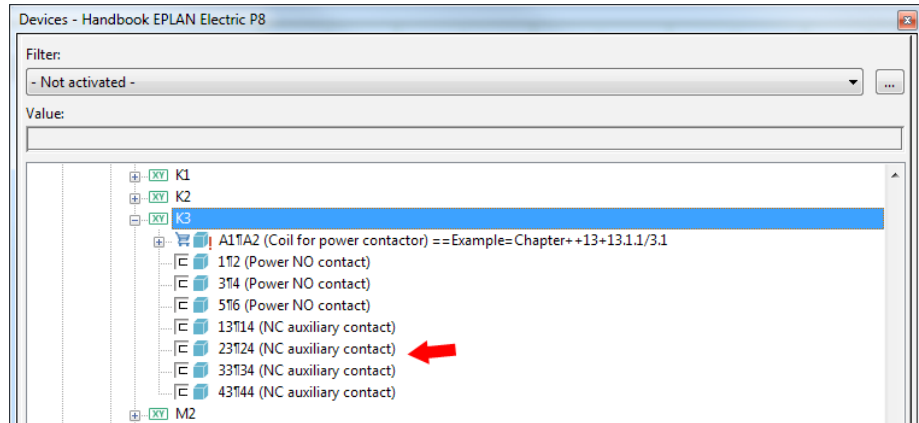


Fig. 13.42
Newly inserted
functions



Question: Can the PDF comment function in EPLAN only be used with Adobe Acrobat Writer (full version)?

Answer: No. You can use, for example, the *redlining* function (reimporting EPLAN comment entries contained in the generated EPLAN-internal PDF) with the freeware PDF-XChange Viewer (status from September 2010).



NOTE: The write protection for the internal EPLAN PDF output should be disabled (**OPTIONS/SETTINGS/USER/INTERFACES/PDF EXPORT** followed by the *General* tab). Otherwise, there could be problems when importing the commented PDF (cannot be imported). This also applies to the Acrobat software solution.



Question: What is the fastest way to find a specific structure identifier in the project?

Answer: Use the F12 – standard shortcut key to launch the **page navigator**. In the navigator, use the right mouse button to select the **DETAILED SELECTION** menu item. This opens the **Detailed selection** dialog. In this dialog, a prefilter is created, which is then used to filter the project's structure identifiers on the basis of a set filter scheme.



Question: Why can I not find a specific structure identifier in the project despite an activated prefilter?

Answer: If a structure identifier in the project cannot be located graphically via an activated prefilter in the **page navigator**, then it is part of an unplaced object.

Unplaced objects and their structure identifiers can be found, for example, in the **device navigator**. The arrangement/sequence of the structure identifiers is imported from the project settings. (under **DISPLAY/PROJECT STRUCTURE (NAVIGATORS)**).



Question: How can lowercase letters be used/entered for a DT designation?

Answer: To be able to also enter lowercase letters for a DT, you need to open **OPTIONS/SETTINGS/PROJECTS/[PROJECT NAME]/DEVICES/DT** and deactivate (remove check mark) the *Conversion to uppercase* setting.